Printer Dover

Spruce version 11.2 -- spooler version 11.2

File: DispY-Rev-Ci.sil etc.

Creation date: 8-Apr-81 13:50:09

For: Spruce

33 total sheets = 32 pages, 1 copy.

Problems encountered:
Font HELVETICA24B substituted for font HELVE24.
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DORADO SCHEMATICS

Display Y

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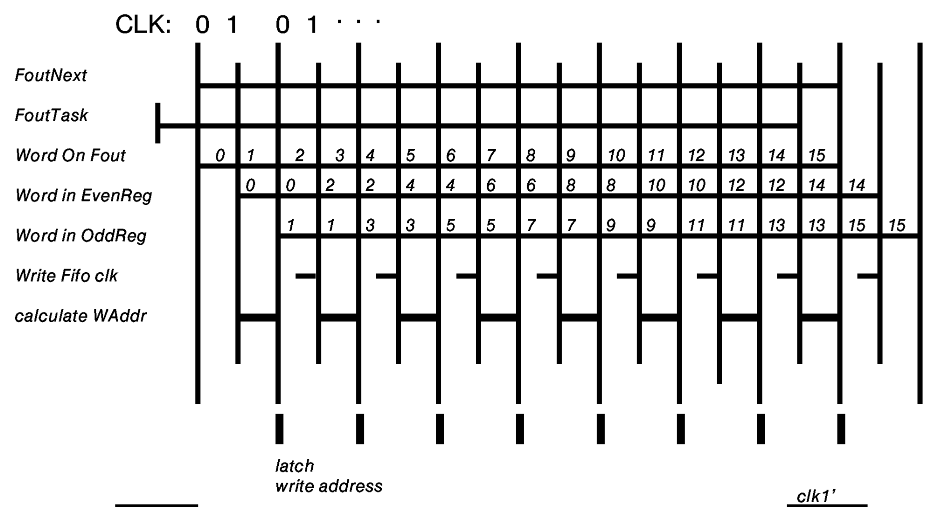

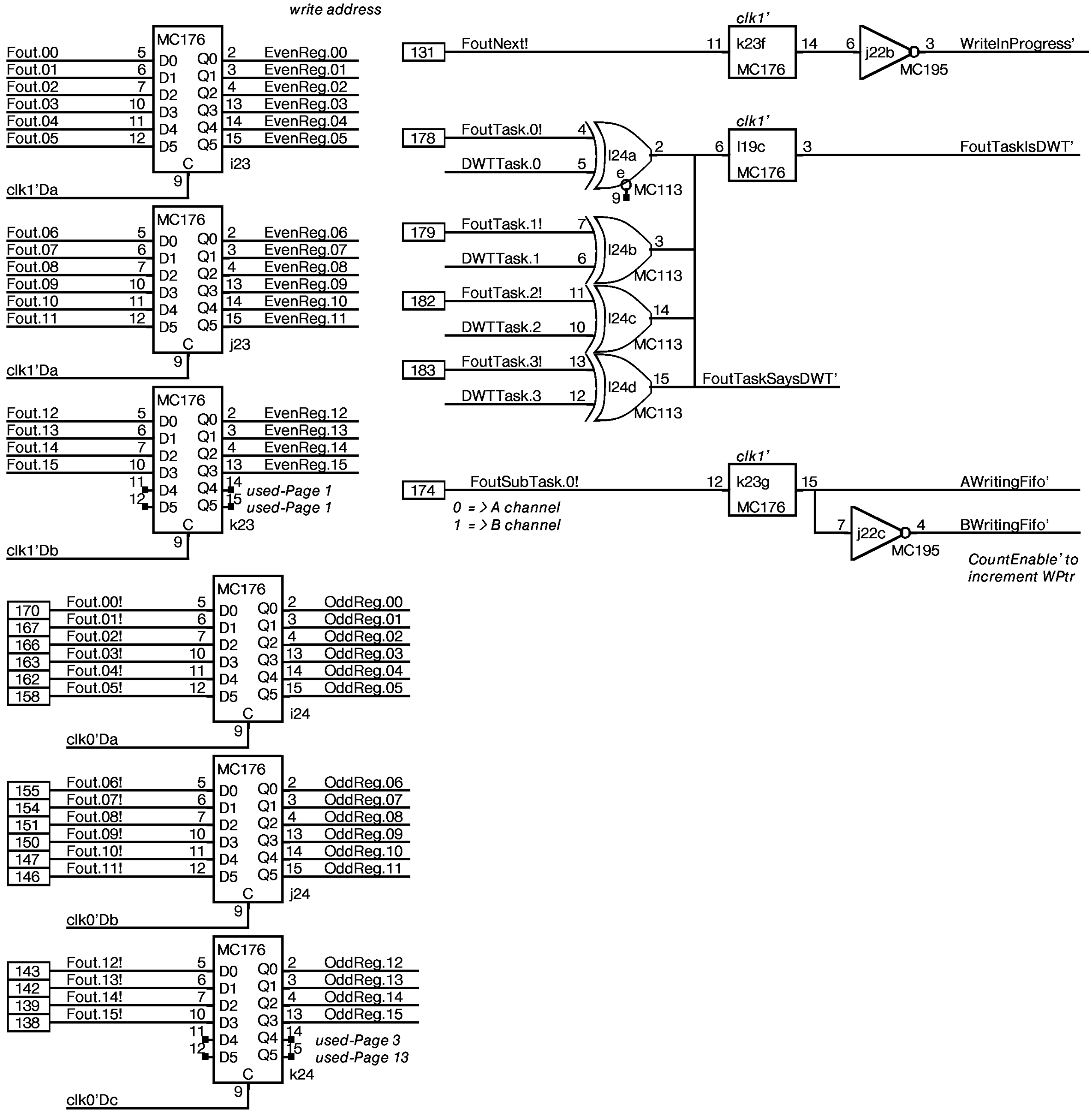

XEROX PARC	Project		File	Designer	Rev	Date	Page
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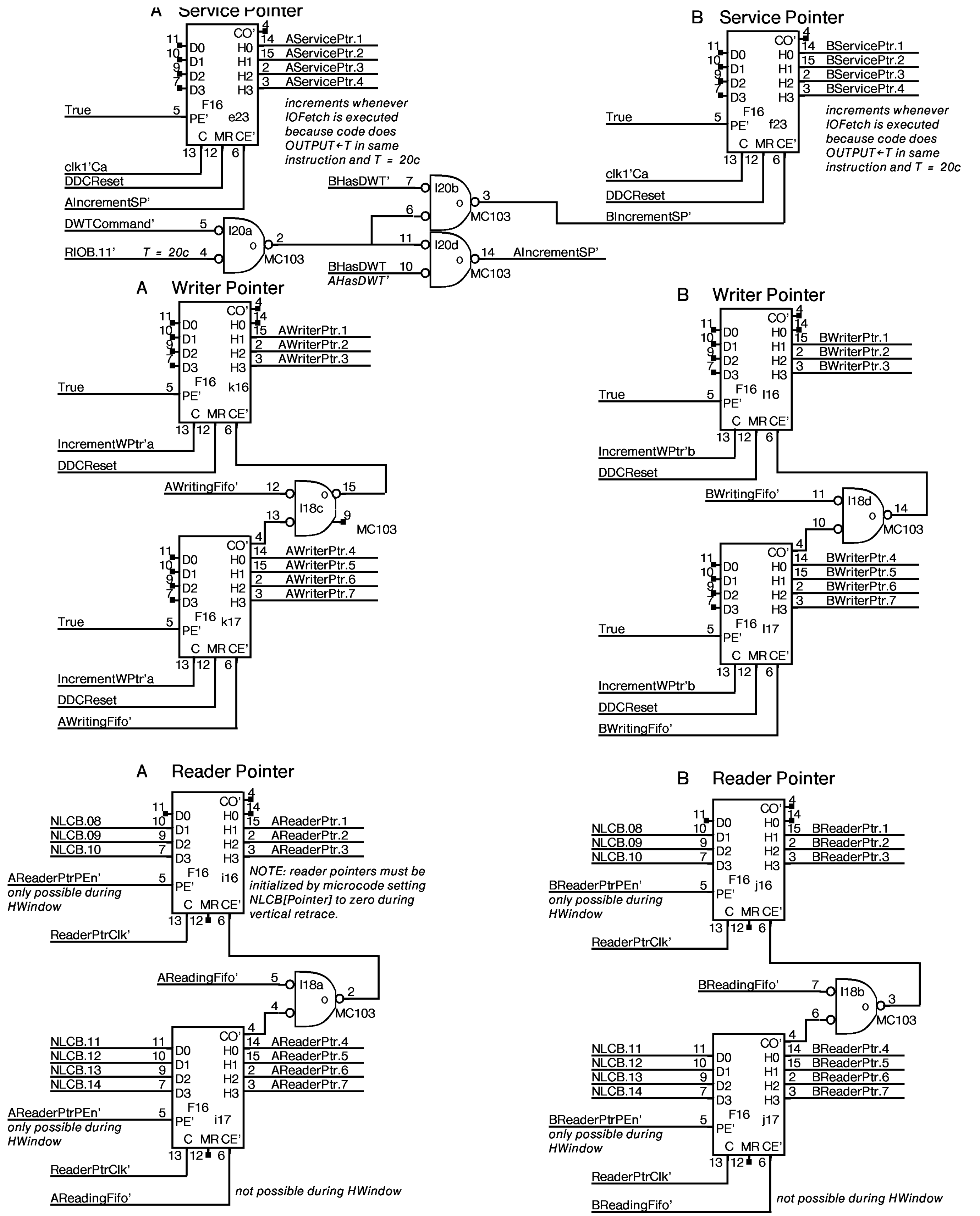
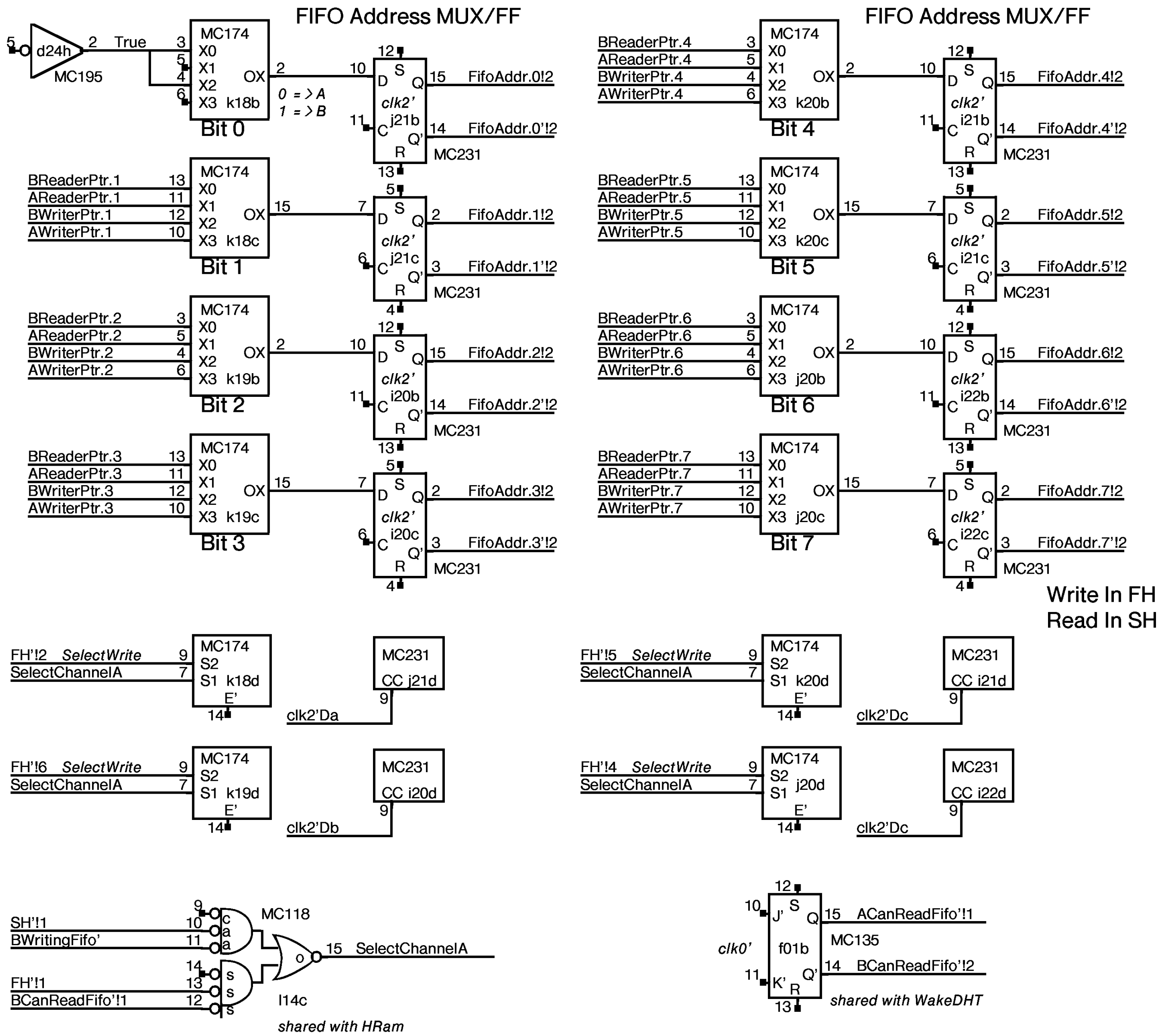
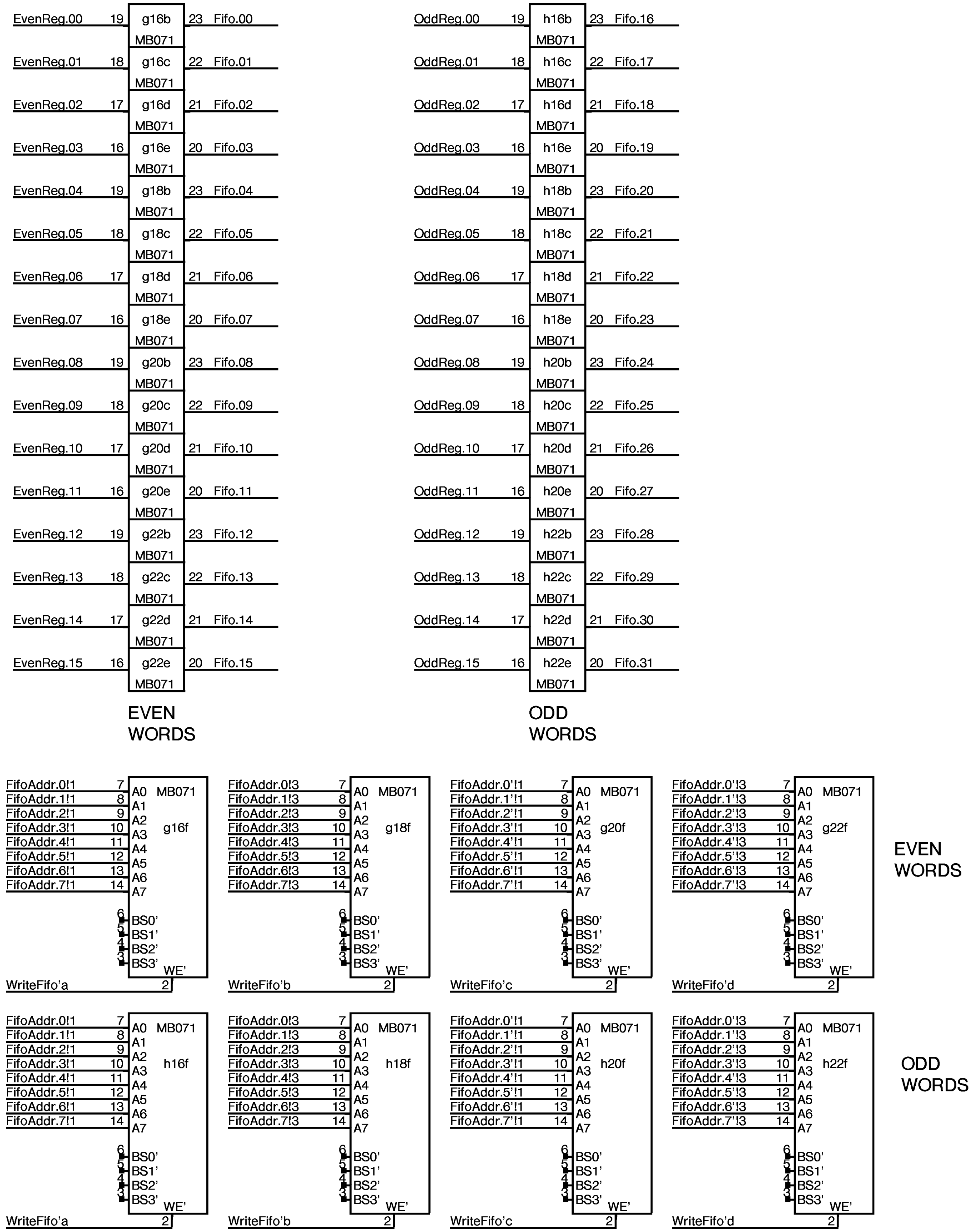


XEROX PARC	Project		File	Designer	Rev	Date	Page
	Dorado	DDC Pointers Reader,Writer,Service	DispY02.sil	K. Pier	Ci	7/19/79	02

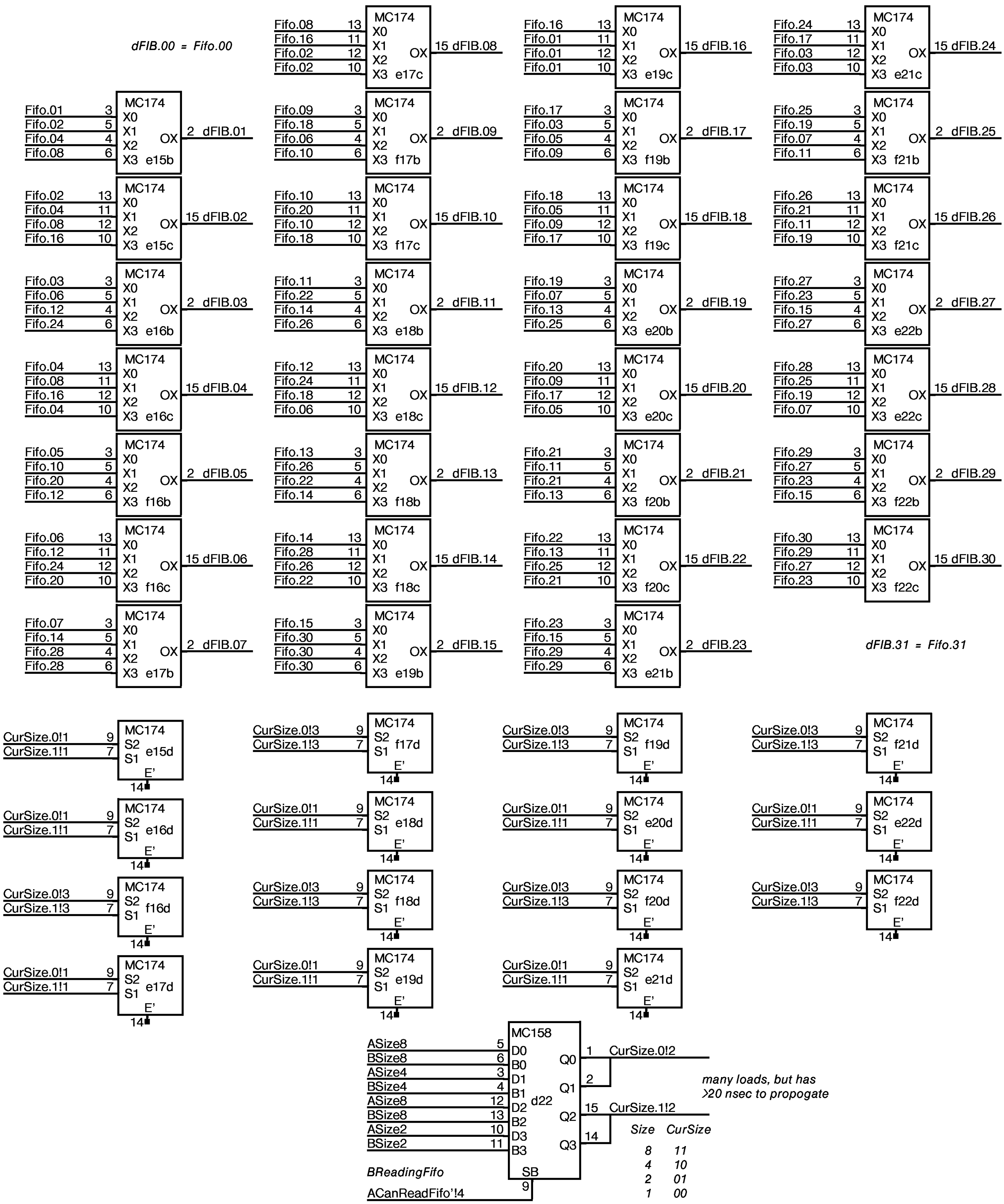


XEROX PARC	Project Dorado	DDC Fifo Address Logic	File DispY03.sil	Designer K. Pier	Rev Ci	Date 11/02/79	Page 03

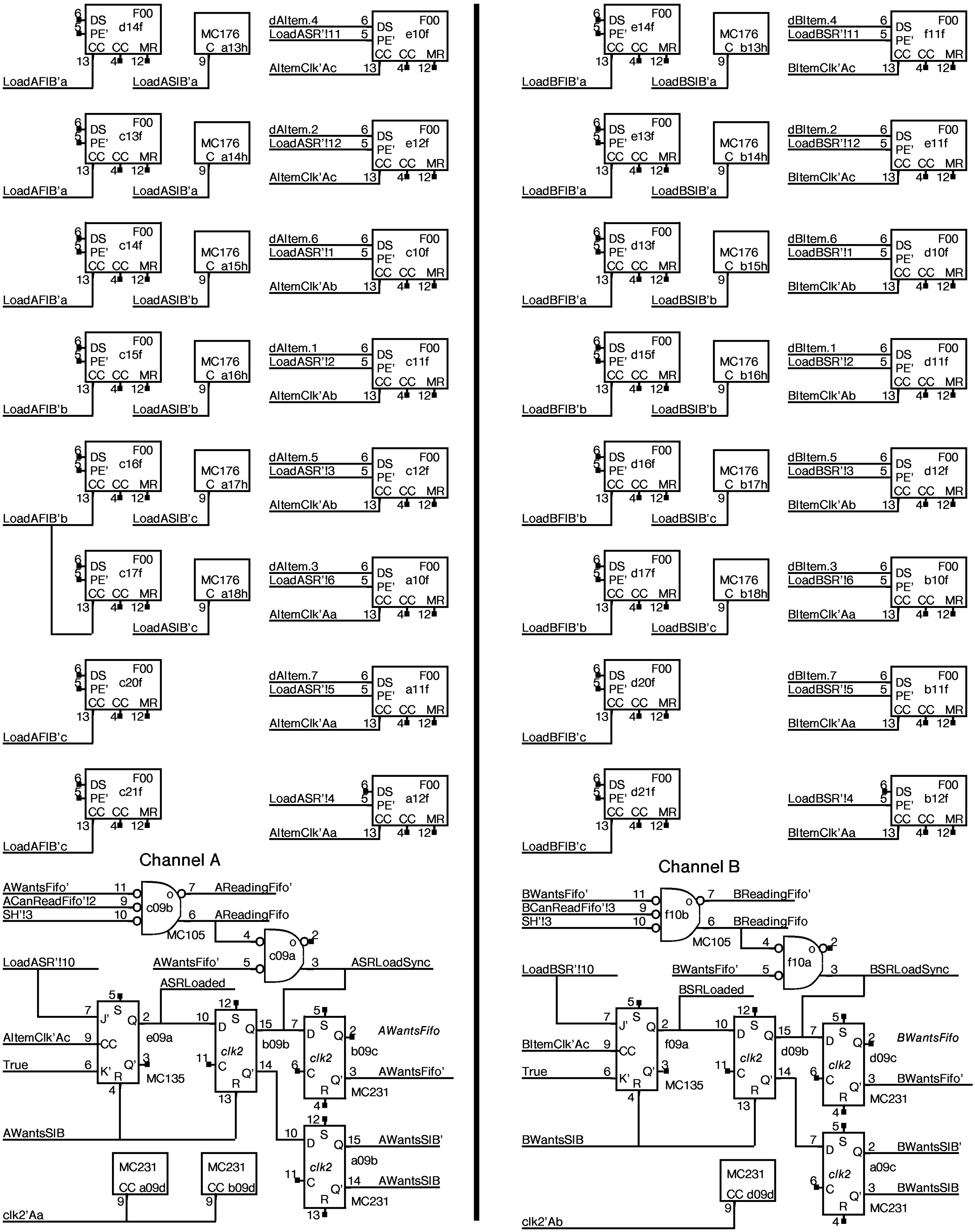


XEROX PARC	Project		File	Designer	Rev	Date	Page
	Dorado	DDC Fifo RAM chips	DispY04.sil	K. Pier	Ci	7/19/79	04

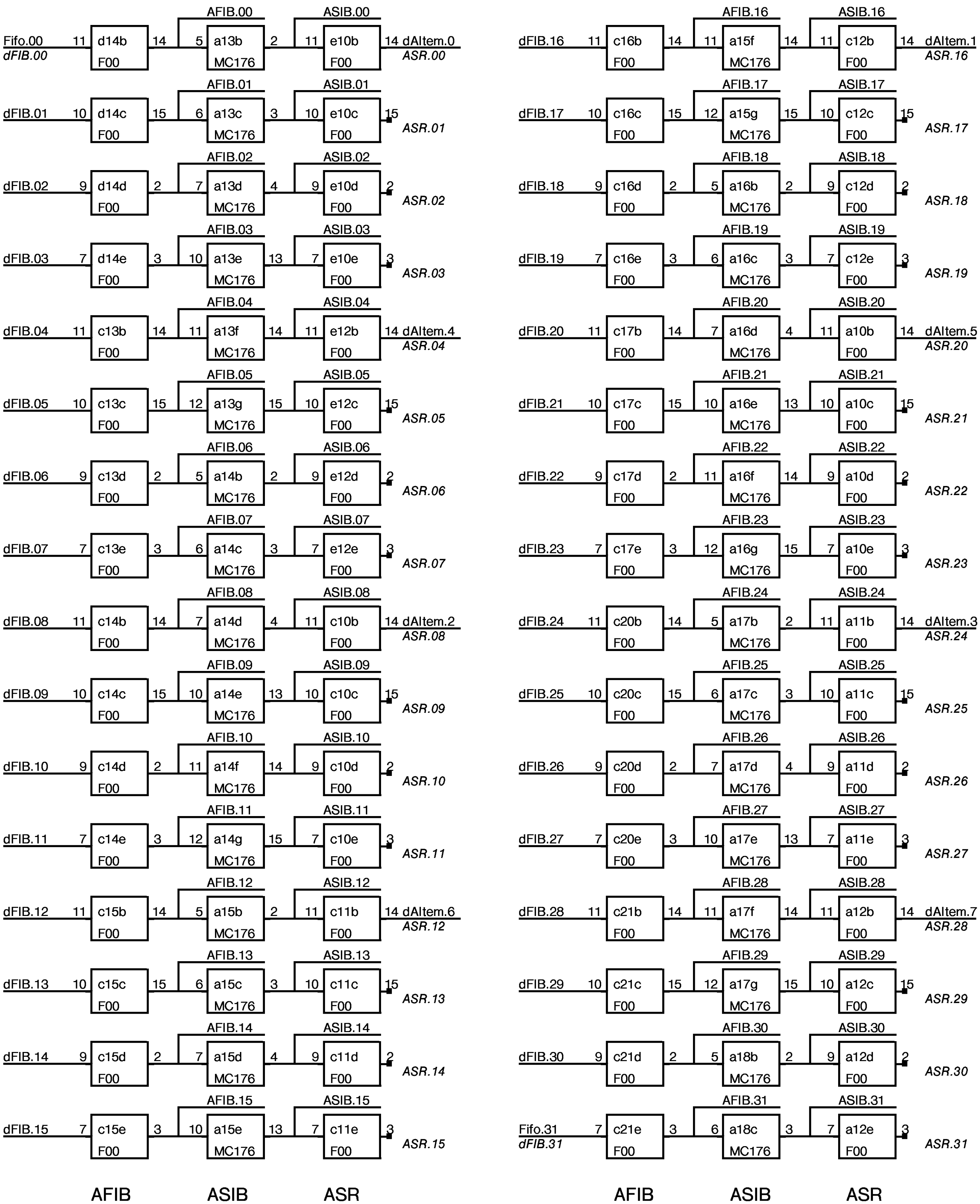
Item Generator Permuter
See ItemGenerator table for input specs



XEROX PARC	Project Dorado	DDC Item Permutation Logic	File DispY05.sil	Designer K. Pier	Rev Ci	Date 11/02/79	Page 05



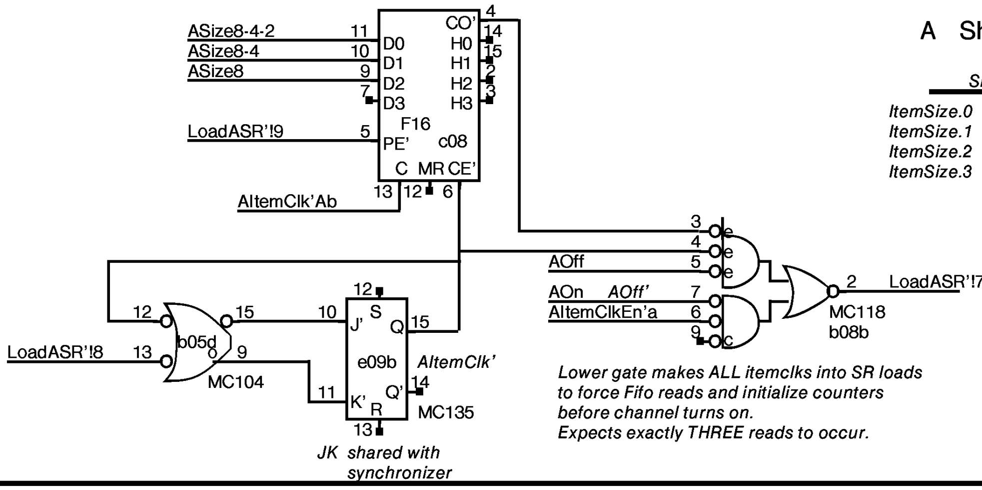
XEROX PARC	Project Dorado	DDC Intermediate Buffer Control	File DispY06.sil	Designer K. Pier	Rev Ci	Date 11/02/79	Page 06



XEROX PARC	Project Dorado	DDC Channel A Data Path	File DispY07.sil	Designer K. Pier	Rev Ci	Date 7/19/79	Page 07

A Shift Register Control

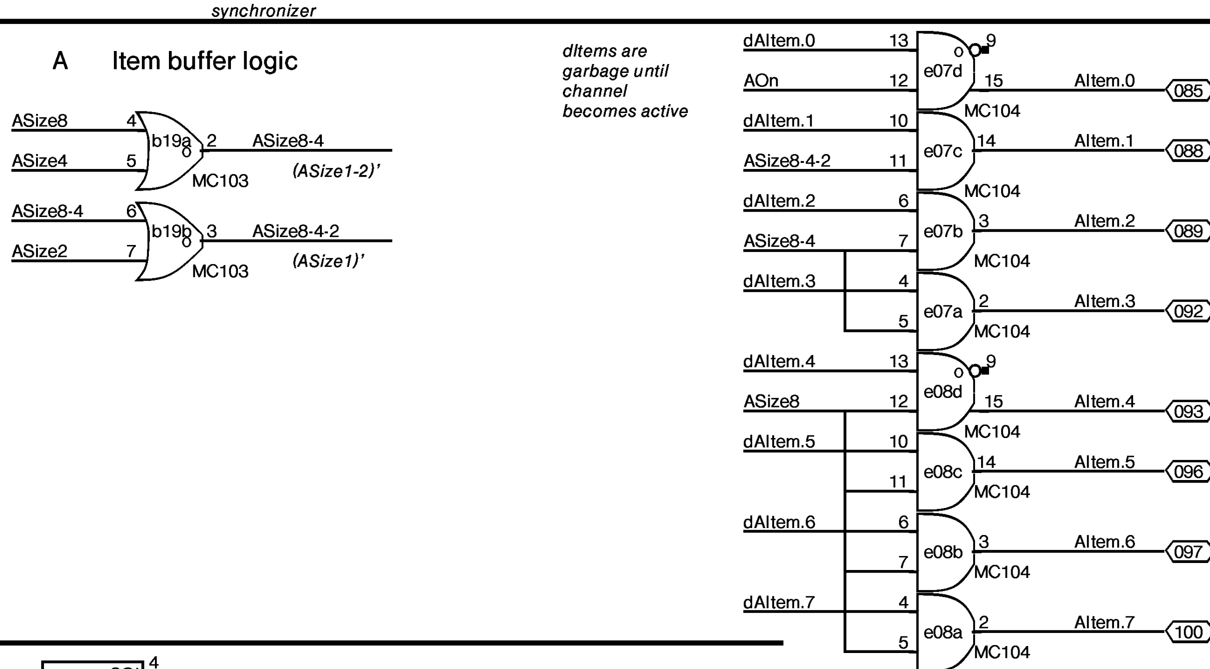
Size	8	4	2	1
ItemSize.0	1	1	1	0
ItemSize.1	1	1	0	0
ItemSize.2	1	0	0	0
ItemSize.3	0	0	0	0



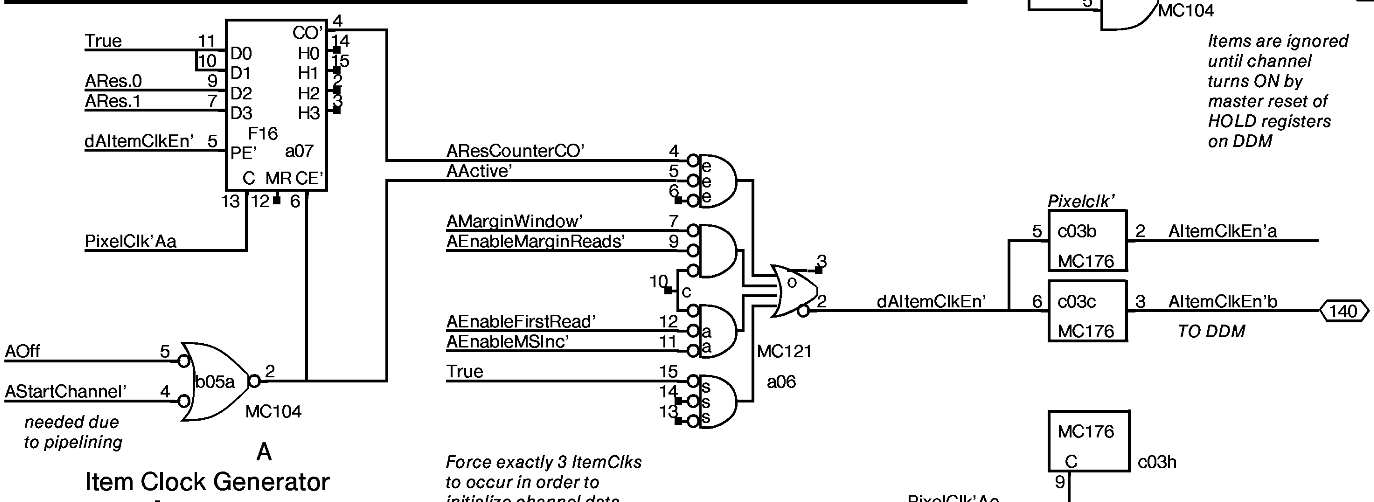
Lower gate makes ALL itemclks into SR loads to force Fifo reads and initialize counters before channel turns on. Expects exactly THREE reads to occur.

A Item buffer logic

dItems are garbage until channel becomes active



Items are ignored until channel turns ON by master reset of HOLD registers on DDM

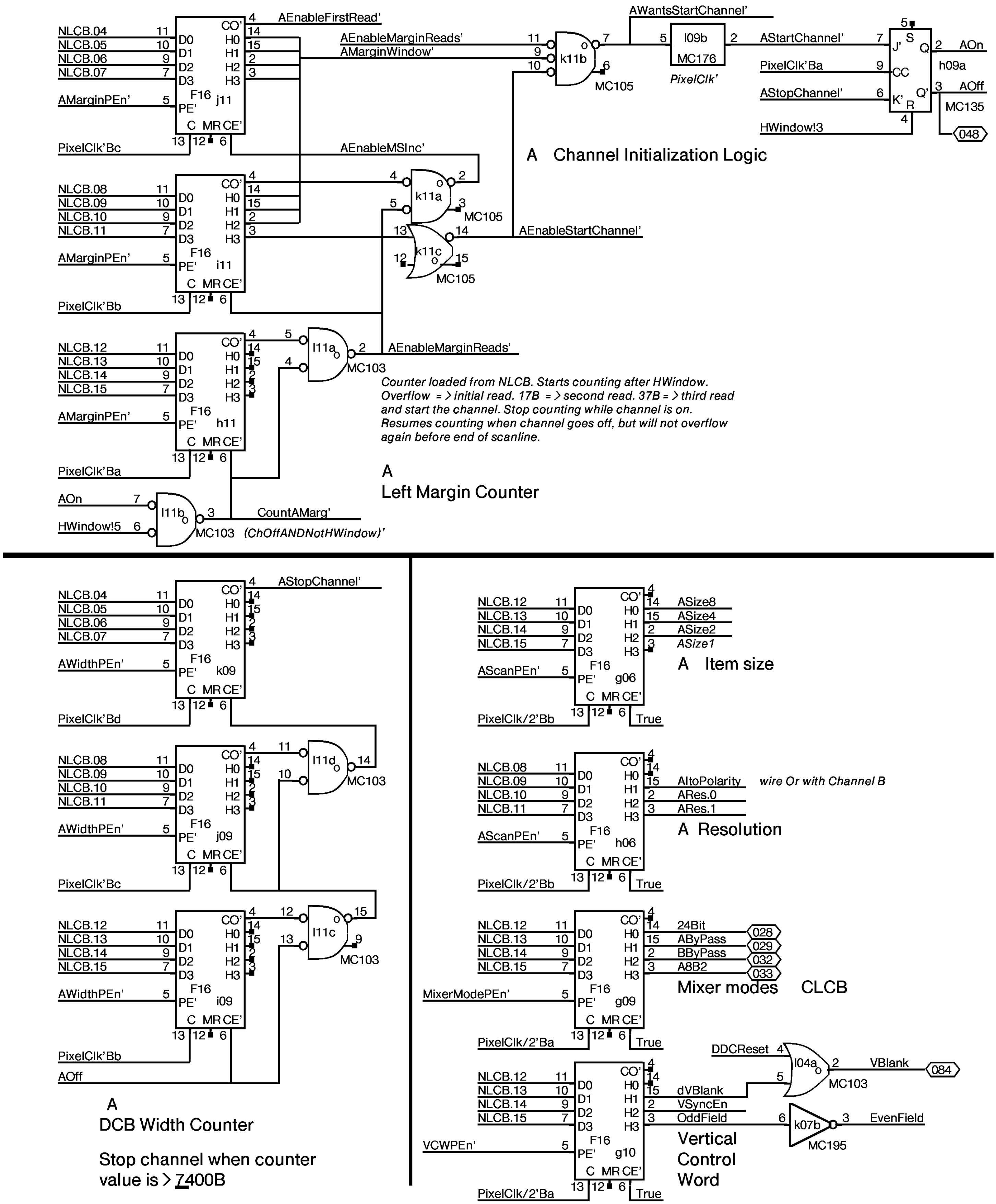


A Item Clock Generator

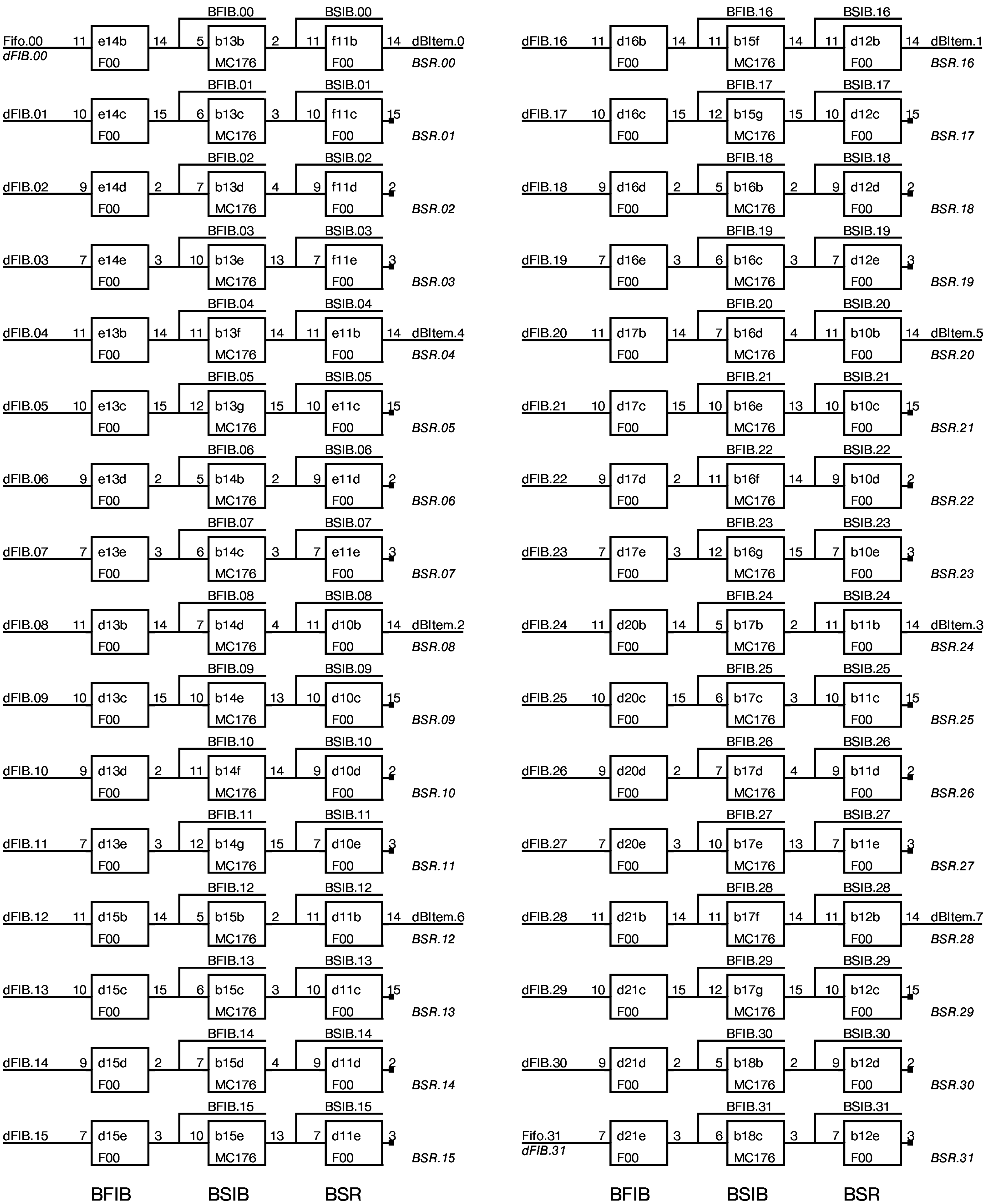
Resolution	Full	Half	Quarter
True	1	1	1
True	1	1	1
Res.0	1	1	0
Res.1	1	0	0

Force exactly 3 ItemClks to occur in order to initialize channel data and clk counters.

The third forced ItemClk loads SR with good data and turns the channel ON.



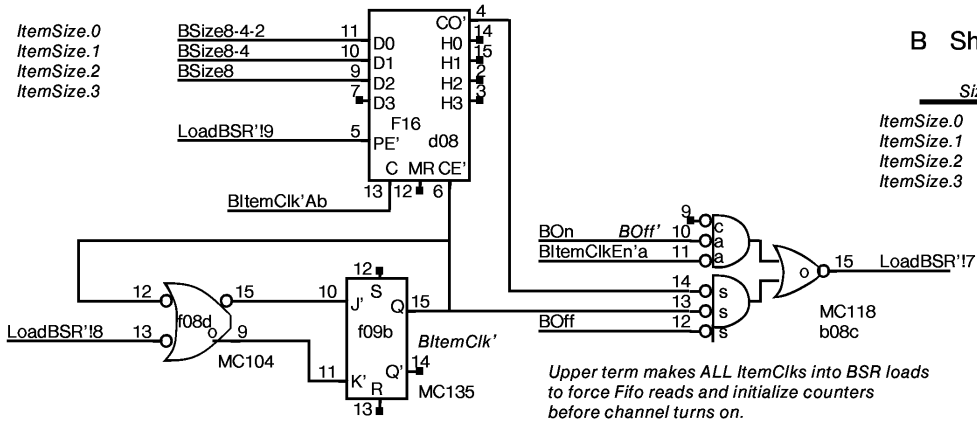
XEROX PARC	Project: Dorado	DDC Channel A Current Line Ctrl Block	File: DispY09.sil	Designer: K. Pier	Rev: Ci	Date: 7/19/79	Page: 09



XEROX PARC	Project Dorado	DDC Channel B Data Path	File DispY10.sil	Designer K. Pier	Rev Ci	Date 7/19/79	Page 10

B Shift Register Control

Size	8	4	2	1
ItemSize.0	*1*	*1*	*1*	*0*
ItemSize.1	*1*	*1*	*0*	*0*
ItemSize.2	*1*	*0*	*0*	*0*
ItemSize.3	*0*	*0*	*0*	*0*

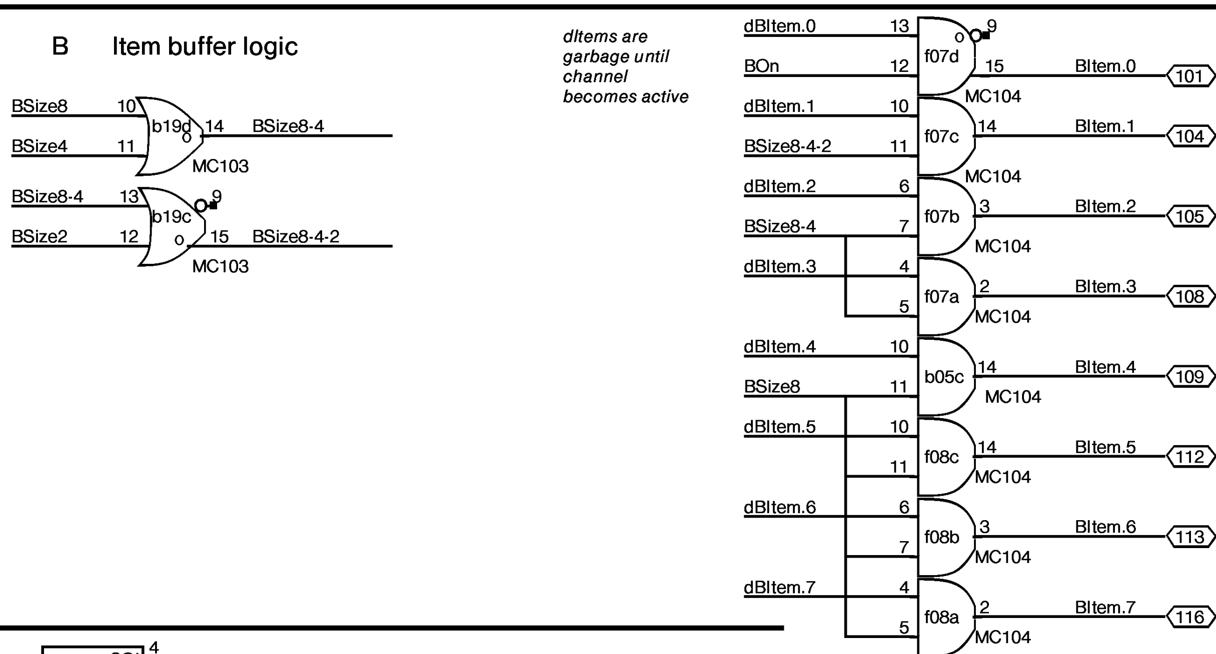


Upper term makes ALL ItemClks into BSR loads to force Fifo reads and initialize counters before channel turns on. Expects exactly THREE reads to occur.

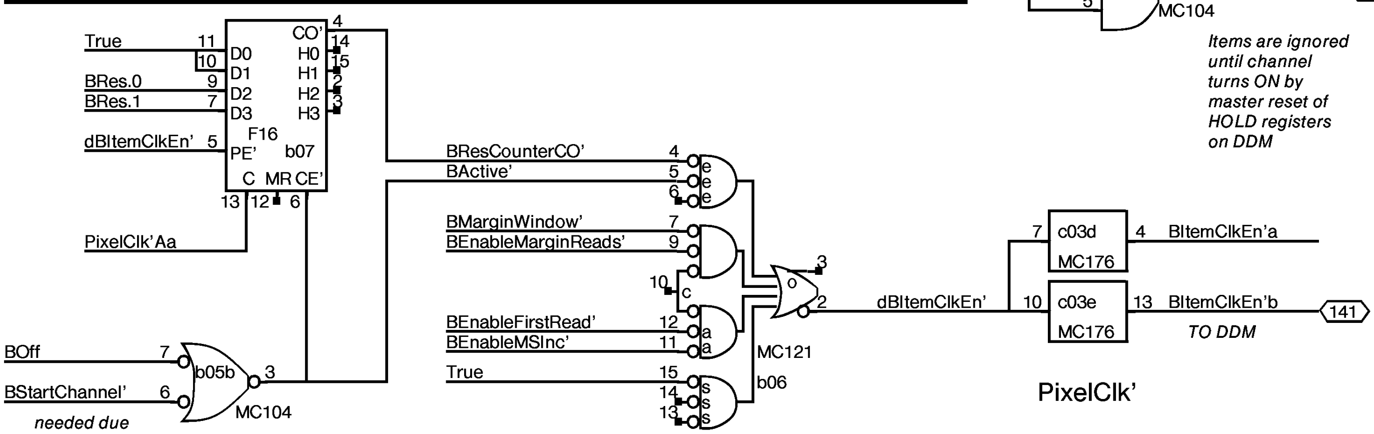
JK shared with synchronizer

B Item buffer logic

dItems are garbage until channel becomes active



Items are ignored until channel turns ON by master reset of HOLD registers on DDM



needed due to pipelining

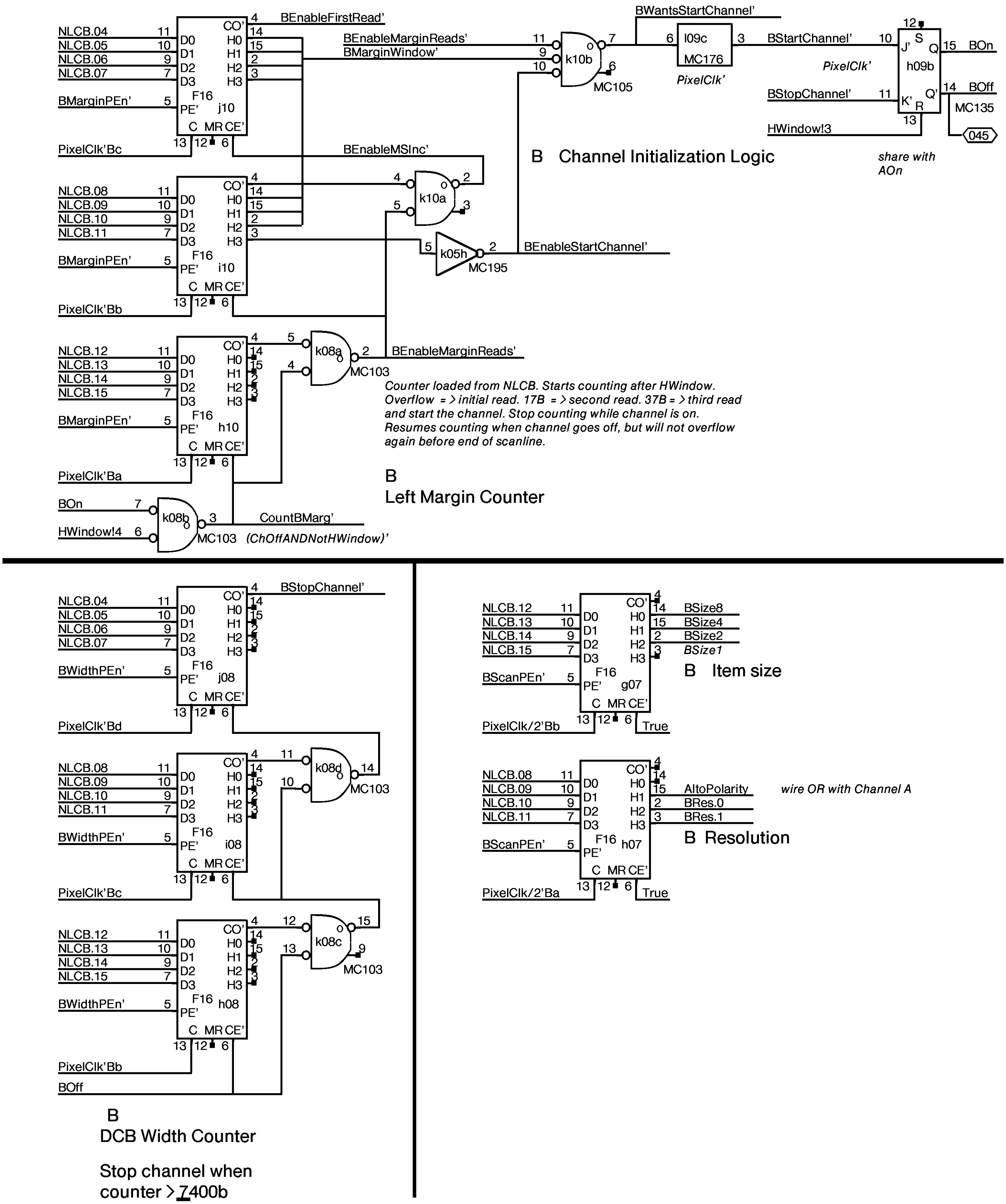
B
Item Clock Generator

Resolution	Full	Half	Quarter
True	*1*	*1*	*1*
True	*1*	*1*	*1*
Res.0	*1*	*1*	*0*
Res.1	*1*	*0*	*0*

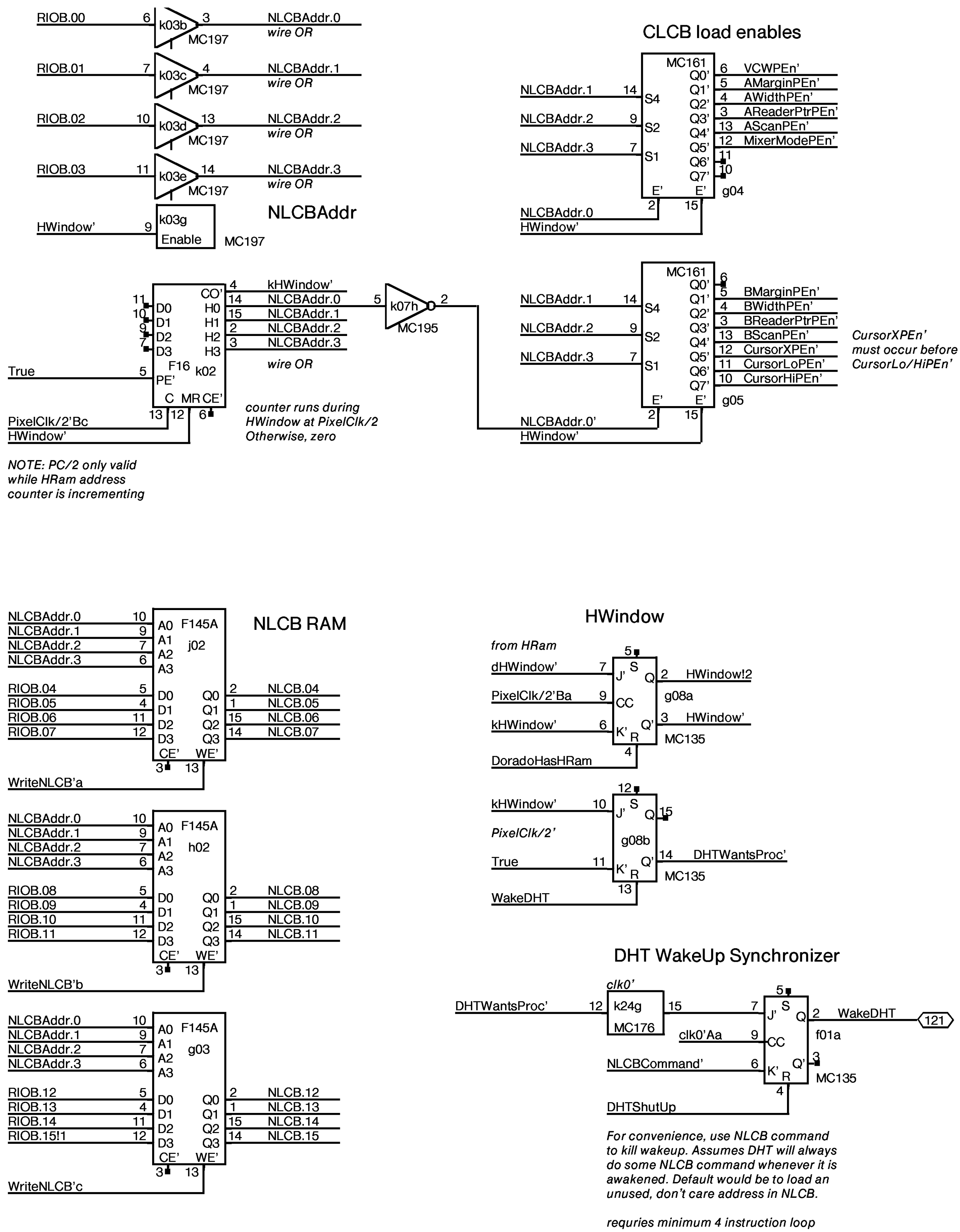
Force exactly 3 ItemClks to occur in order to initialize channel data and clk counters.

The third forced ItemClk loads SR with good data and turns the channel ON.

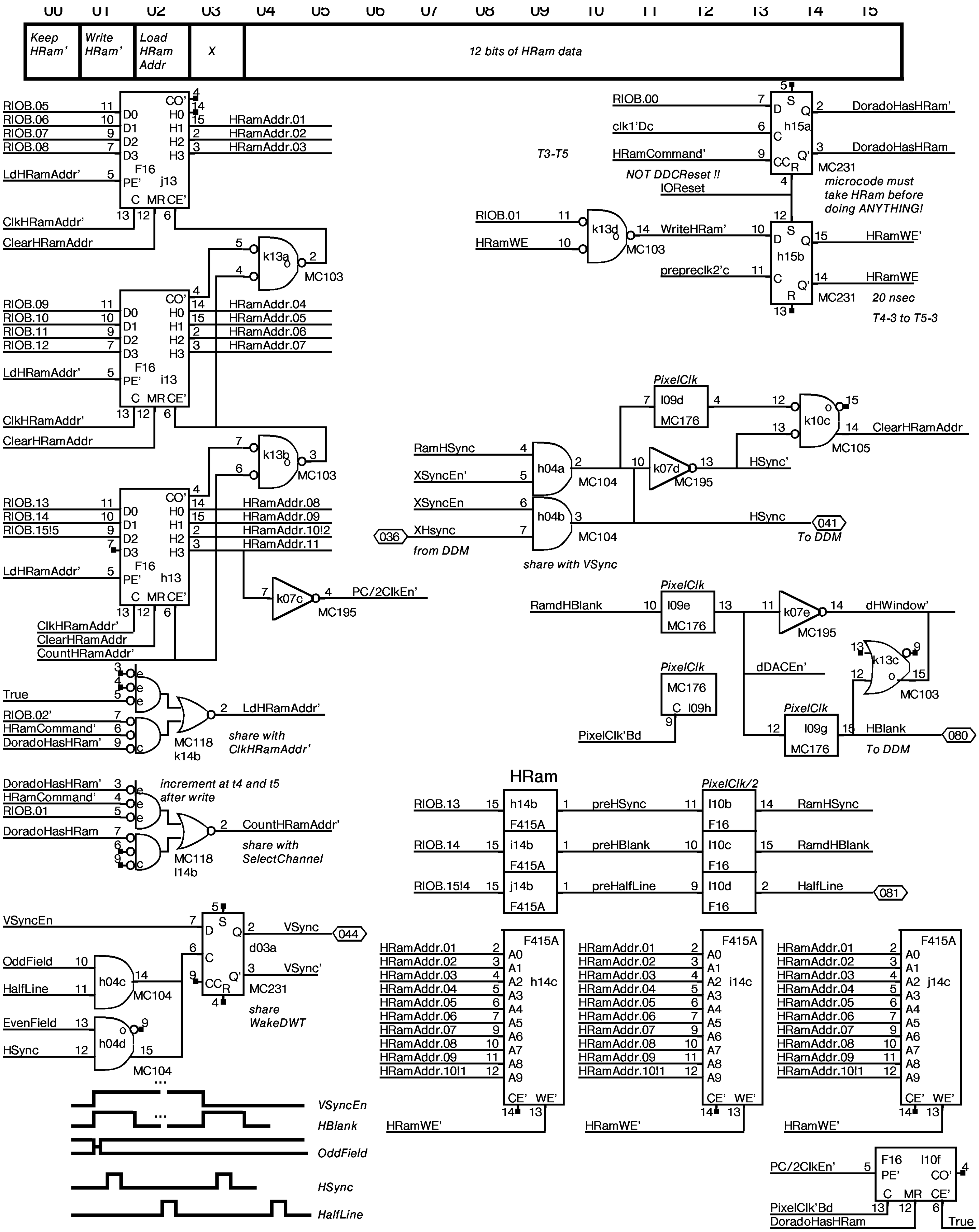
XEROX PARC	Project: Dorado	DDC Channel B Items, SR and ItemClk Control	File: DispY11.sil	Designer: K. Pier	Rev: Ci	Date: 7/19/79	Page: 11



XEROX PARC	Project Dorado	DDC Channel B Current Line Ctrl Block	File DispY12.sil	Designer K. Pier	Rev Ci	Date 7/19/79	Page 12



XEROX PARC	Project		File	Designer	Rev	Date	Page
	Dorado	DDC Next Line Control Block	DispY13.sil	K. Pier	Ci	7/19/79	13



XEROX PARC	Project Dorado	DDC Horizontal RAM	File DispY14.sil	Designer K. Pier	Rev Ci	Date 7/19/79	Page 14

THE DDC MAINTAINS, FOR EACH CHANNEL, TWO FLAGS:

NEXT Word Control Block Flag (NextWCBFlag)
Current Word Control Block Flag (CurrentWCBFlag)

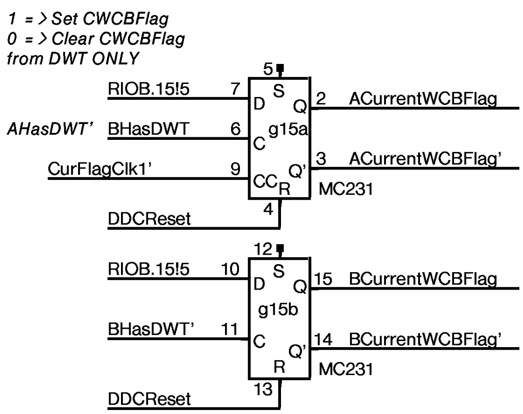
A Word Control Block is a pair of values called Address and MunchCount, for either the CURRENT or the NEXT scanline.

Flag management	SET	CLEARED
NextWCBFlag	*by DHT when it has filled the NextWCB*	*by DWT when it has copied NextWCB into CurrentWCB*
CurrentWCBFlag	*by DWT when it has copied NextWCB into CurrentWCB*	*by DWT after it has sent out all the data from the CurrentWCB*

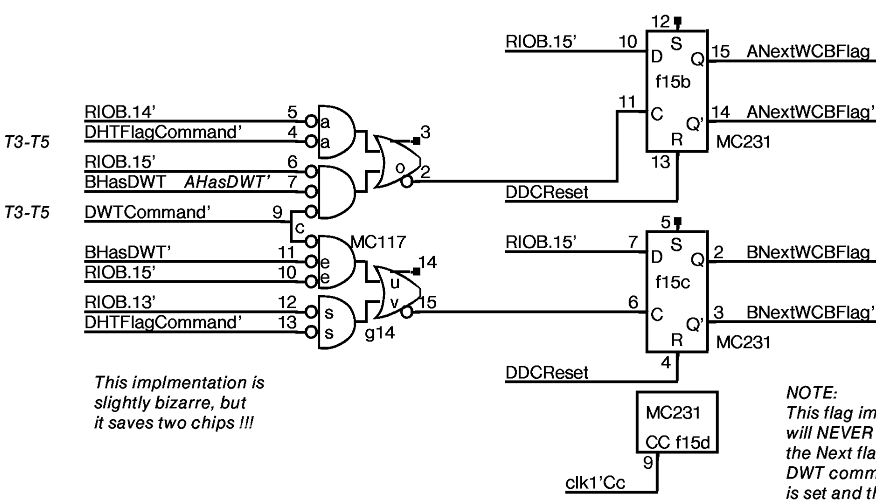
WakeUp conditions:

WakeDHT: whenever CLCB ← NLCB   i.e.: end of every HWindow

WakeDWT: (CurrentWCBFlag AND BufferAvailable)   OR   (NextWCBFlag AND NOT(CurrentWCBFlag))



This circuit ASSUMES:
1. a DWT command with RIOB.11 set is to be ignored.   *See clock page.*
2. Any DWT command with RIOB.11 low is a CWCBFlag command, either set or clear.   *See clock page.*



NOTE:
This flag implementation assumes that DHT flag commands will NEVER set bit 15 and will ALWAYS set either 13 or 14, the Next flag to set.   Also, it assumes the ONLY kind of DWT command it should respond to is one in which bit 15 is set and therefore NextWCBFlag is to be cleared.

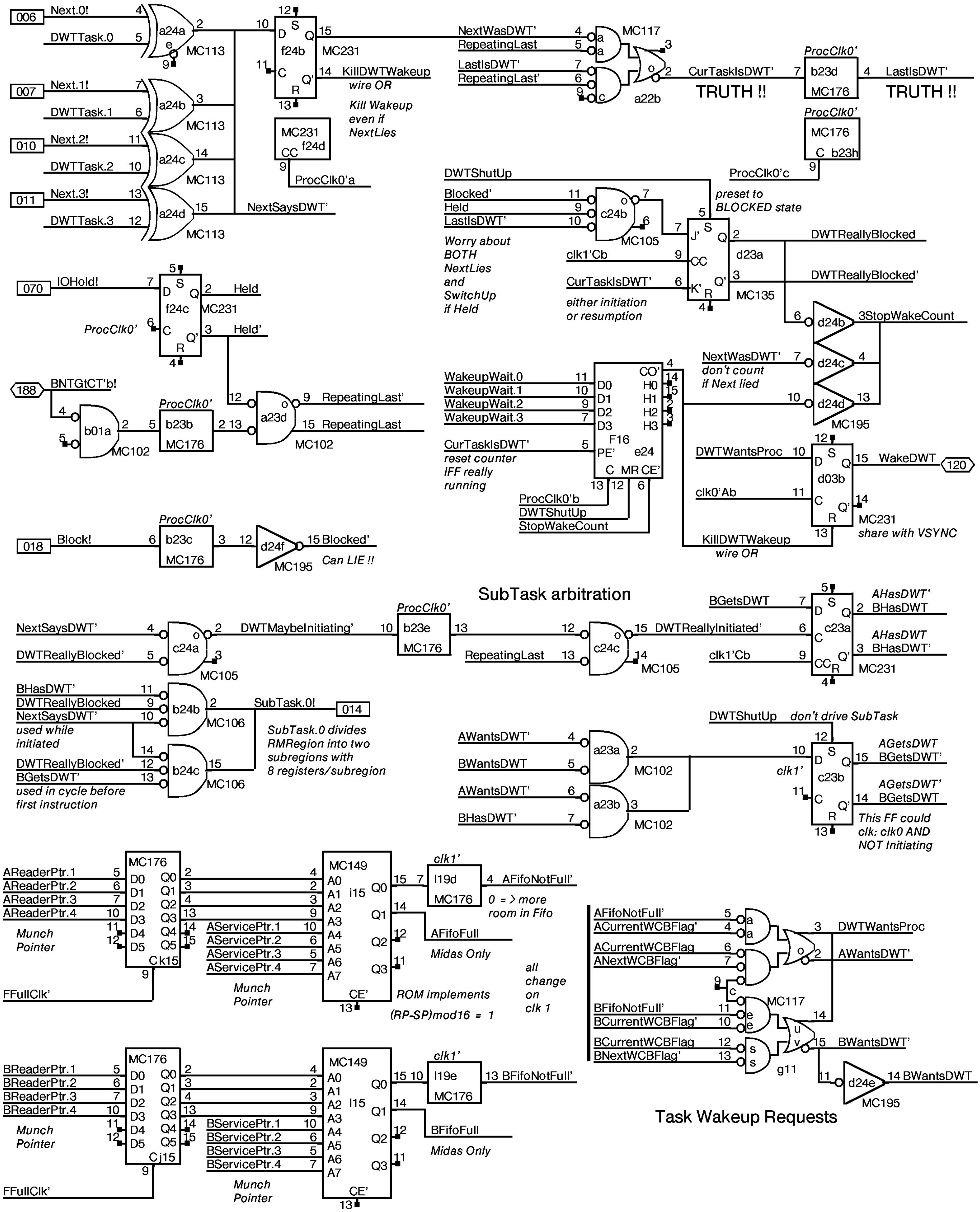
For DWT commands:

RIOB	Means
= 20c	IOFetch signal don't touch Flags
= 1c	Set CWCBFlag and Clear NWCBFlag
= 0c	Clear CWCBFlag

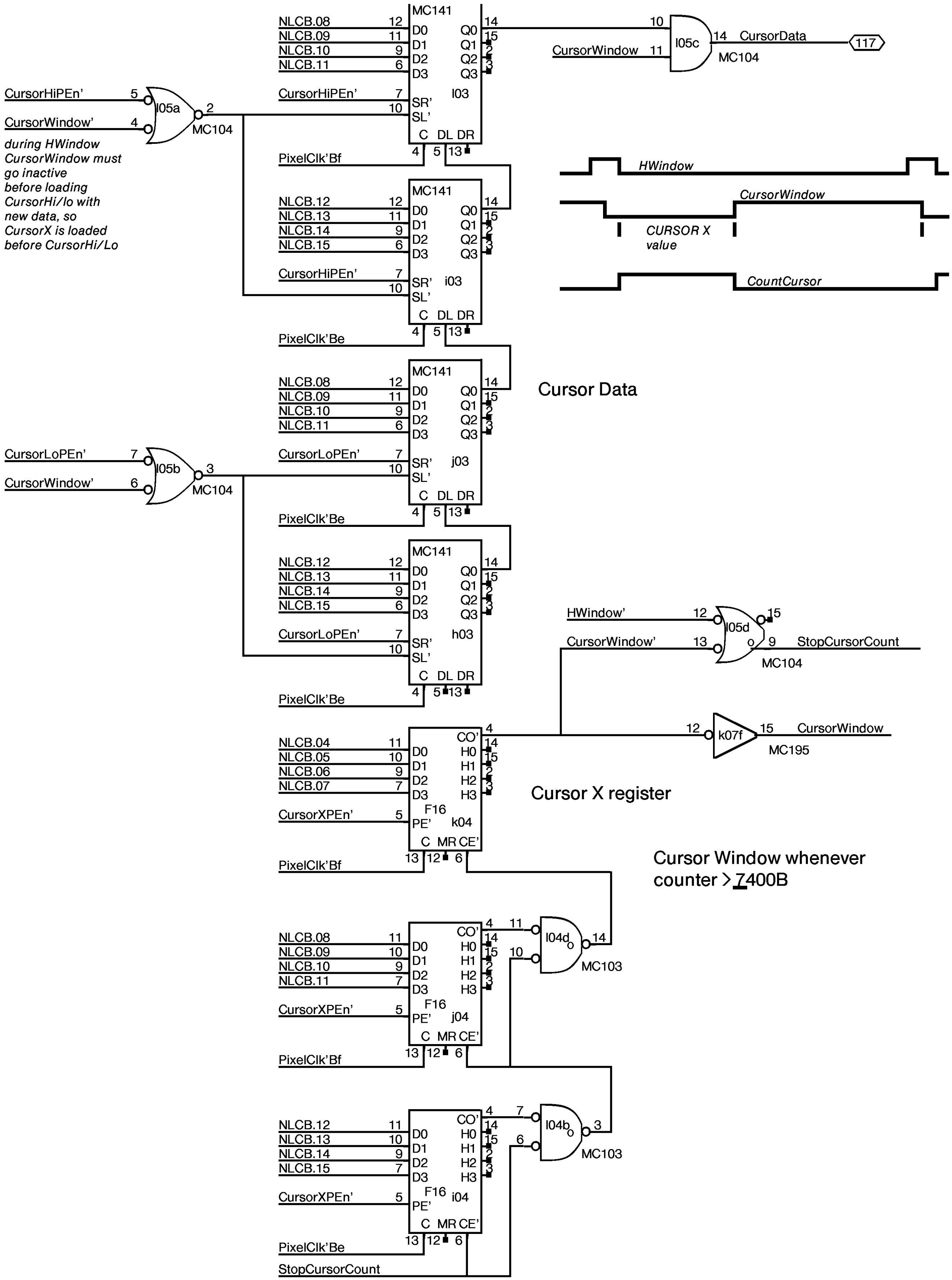
For DHT commands:

RIOB	Means
= 2c	Set ANextWCBFlag
= 4c	Set BNextWCBFlag

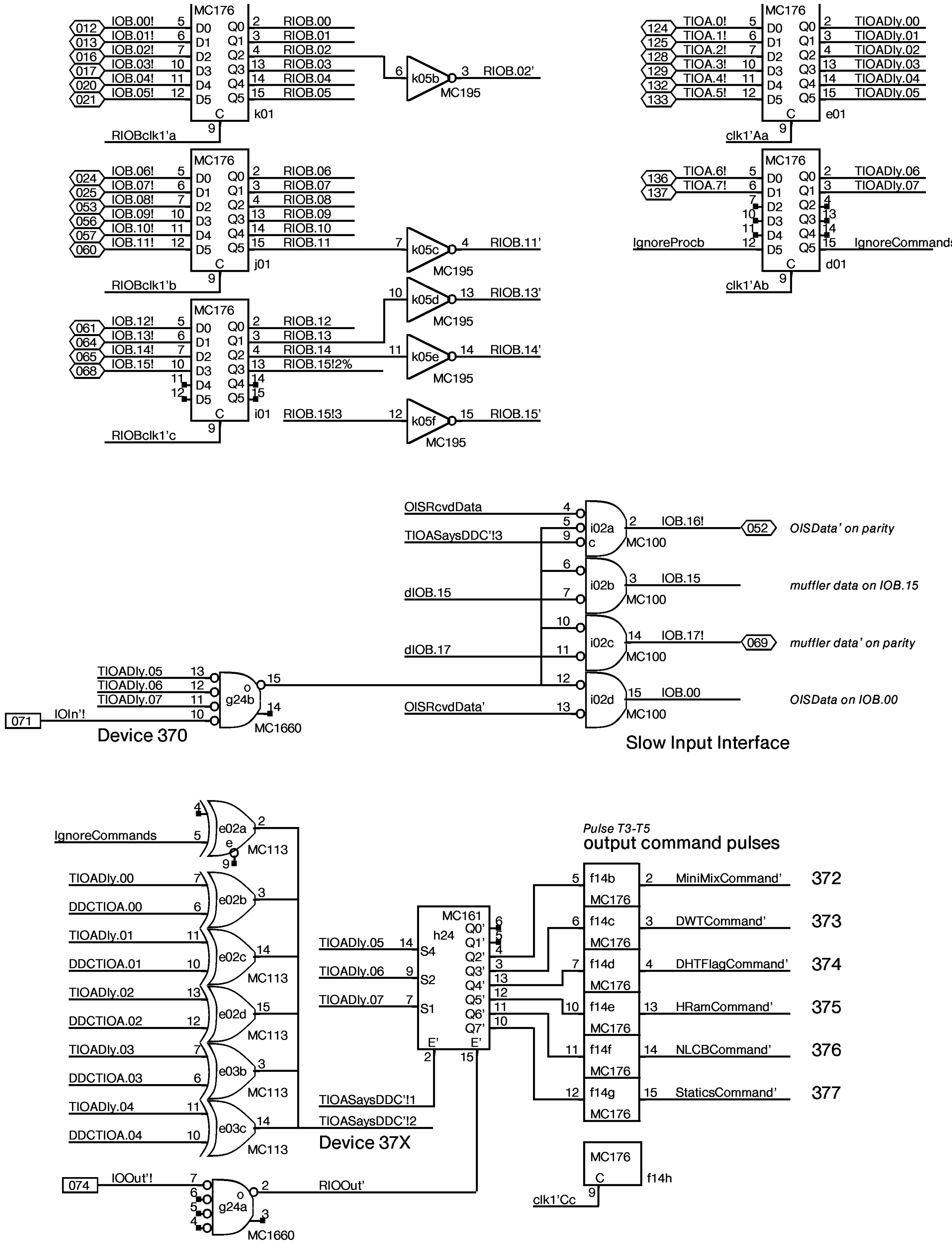
XEROX PARC	Project		File	Designer	Rev	Date	Page
	Dorado	DDC Flag control logic	DispY15.sil	K. Pier	Ci	7/19/79	15



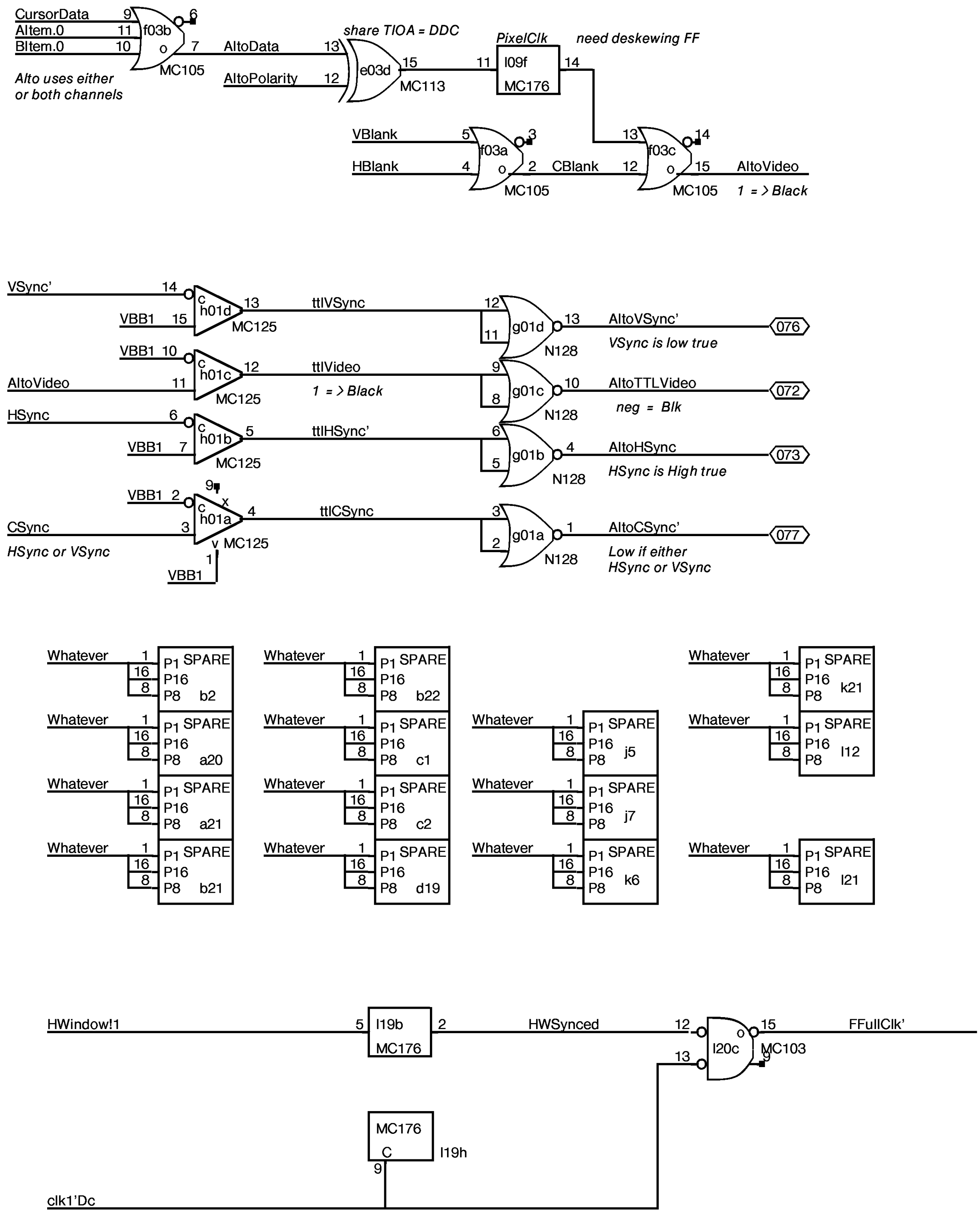
XEROX PARC	Project Dorado	DDC DWT Task Wakeup Logic	File DispY16.sil	Designer K. Pier	Rev Ci	Date 4/07/81	Page 16



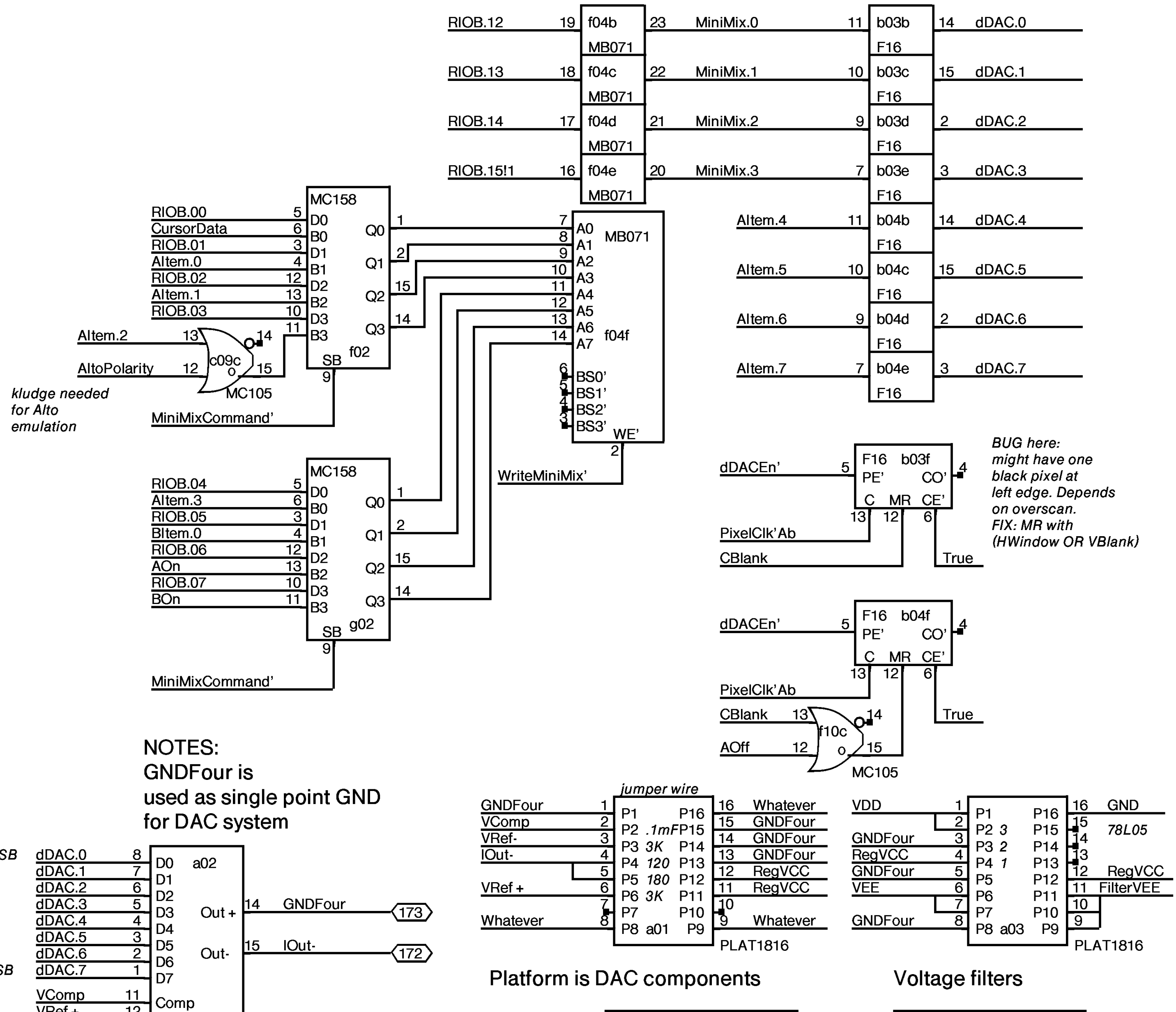
XEROX PARC	Project Dorado	DDC Cursor Hardware	File DispY17.sil	Designer K. Pier	Rev Ci	Date 7/19/79	Page 17



XEROX PARC	Project Dorado	DDC Slow IO Interface	File DispY18.sil	Designer K. Pier	Rev Ci	Date 7/19/79	Page 18



XEROX PARC	Project: Dorado	DDC Alto Display Drivers	File: DispY19.sil	Designer: K. Pier	Rev: Ci	Date: 3/26/81	Page: 19



Digital/Analog converter

Platform is DAC components

Voltage filters

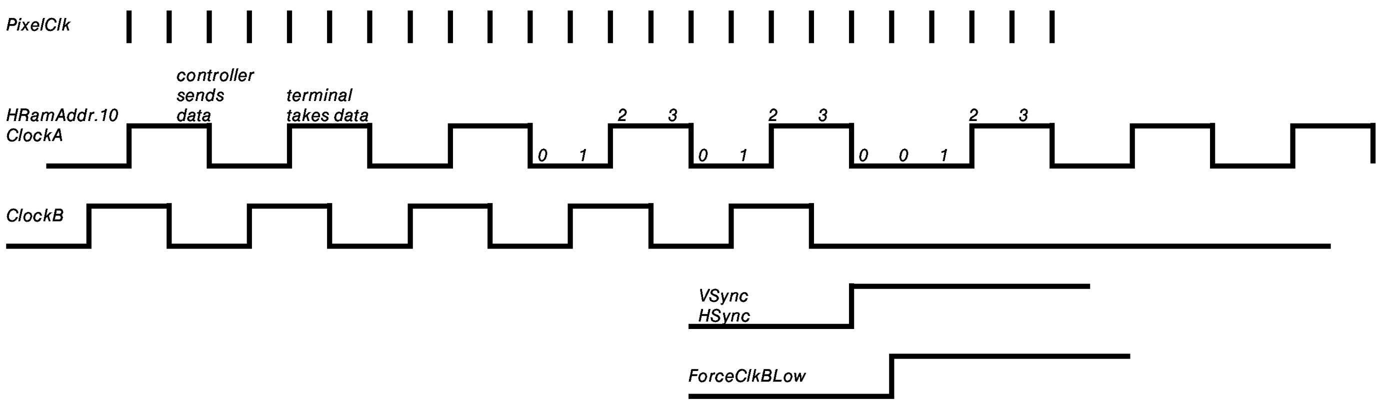
1	jumper wire	16
2	0.1mFarad	15
3	3 KOhm	14
4	120 Ohm	13
5	180 Ohm	12
6	3 KOhm	11
7		10
8		09

location a01

1	0.1mFarad	16
2 in 3	78L05	15
3 gnd 2		14
4 out 1		13
5	0.1mFarad	12
6	12 mHnry	11
7	2 ohm	10
8	+ 22 mF	09

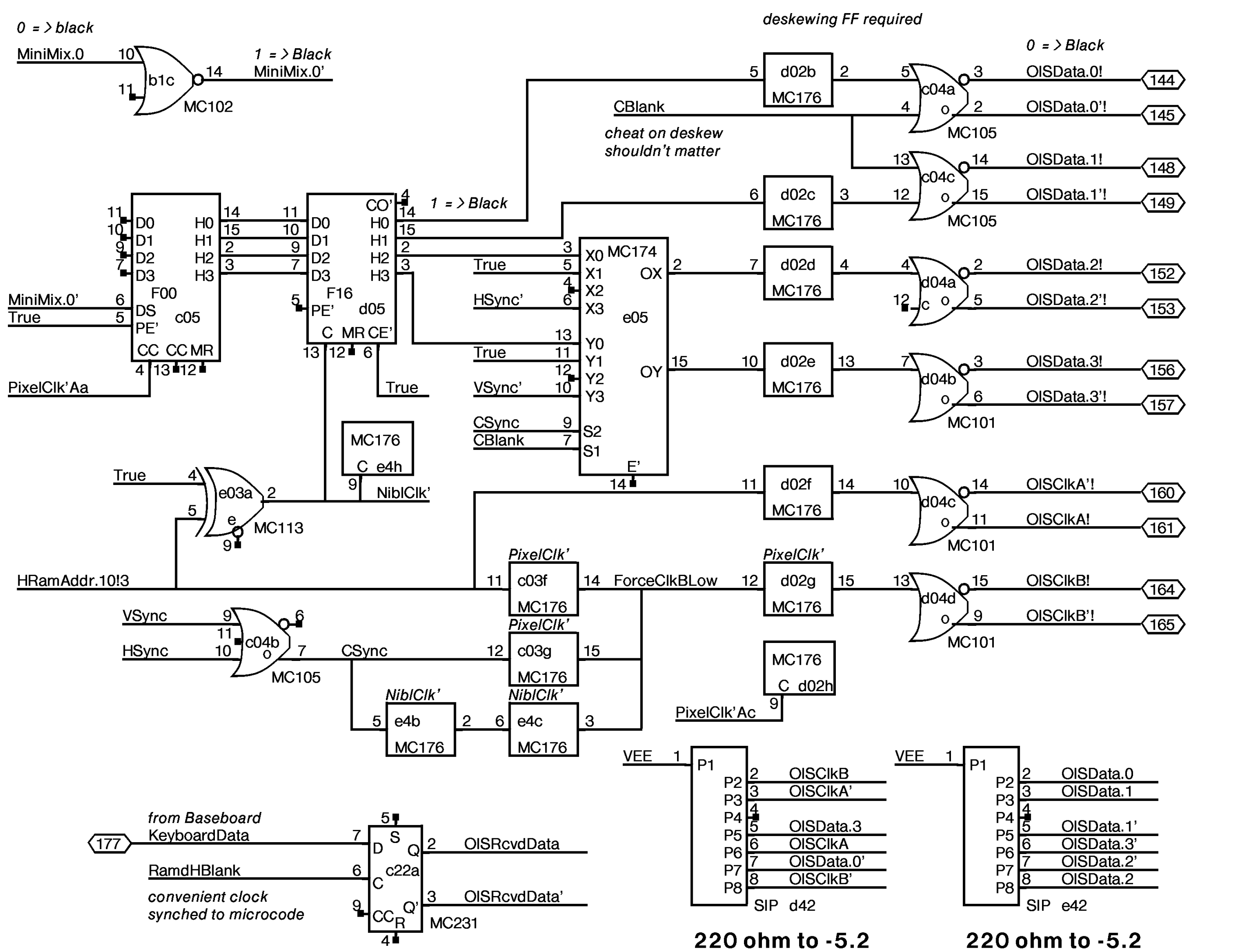
location a03

RevCd: swapped positions of 12mH and 2mF

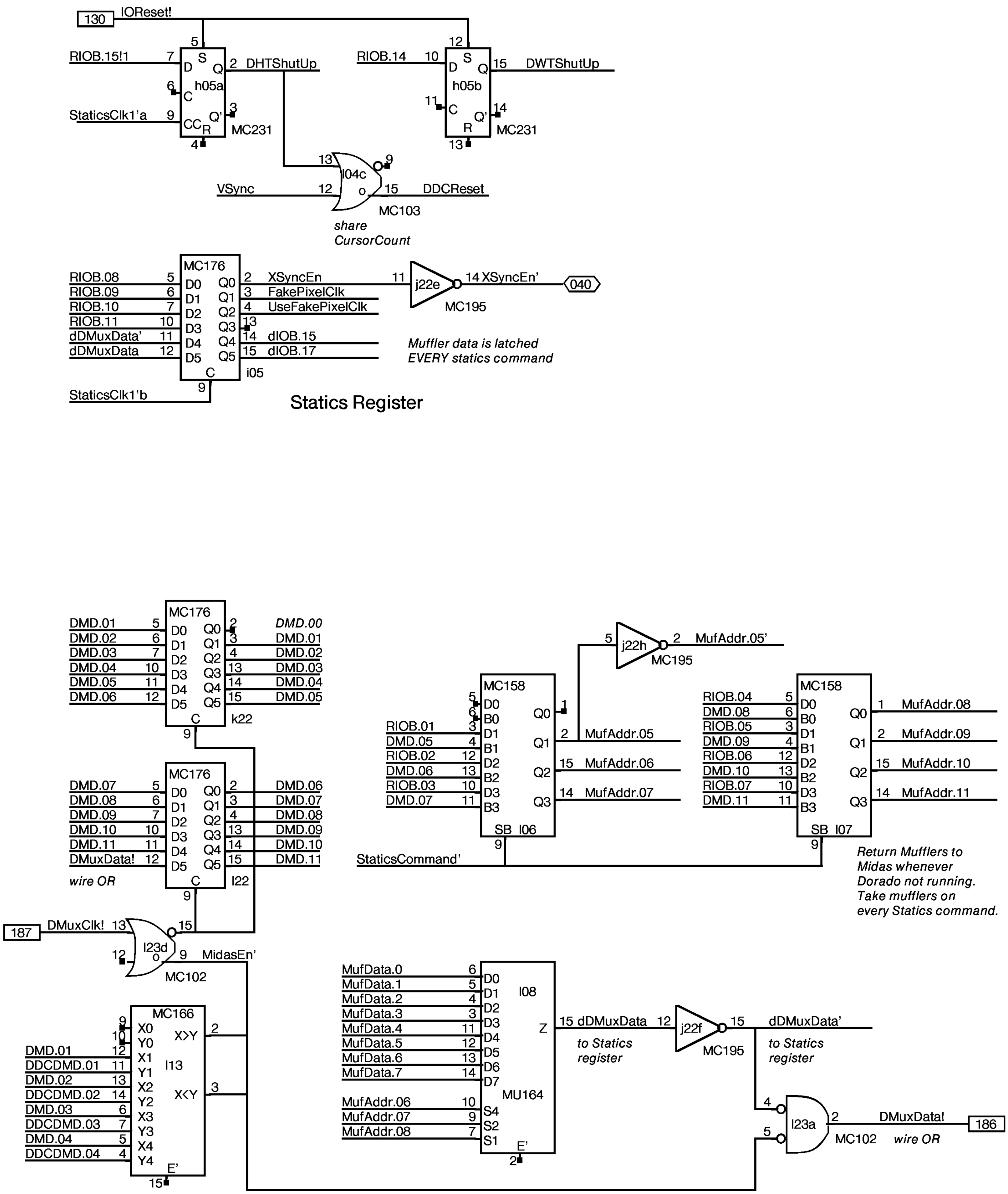


CLAIM: HRamAddr.10 is the waveform for ClockA. Delay it by one pixelclock, and invert, then you get the waveform for ClockB. ClockB is also jammed low by (HSync OR VSync) delayed by one PixelClk. To avoid hiccups in the phasing of ClockA with respect to ClockB, HSync should only be generated just after HRamAddr.10-11 = 3, that is, a hiccup would occur if HRamAddr.01-11 = 1 just before HSync occurred.

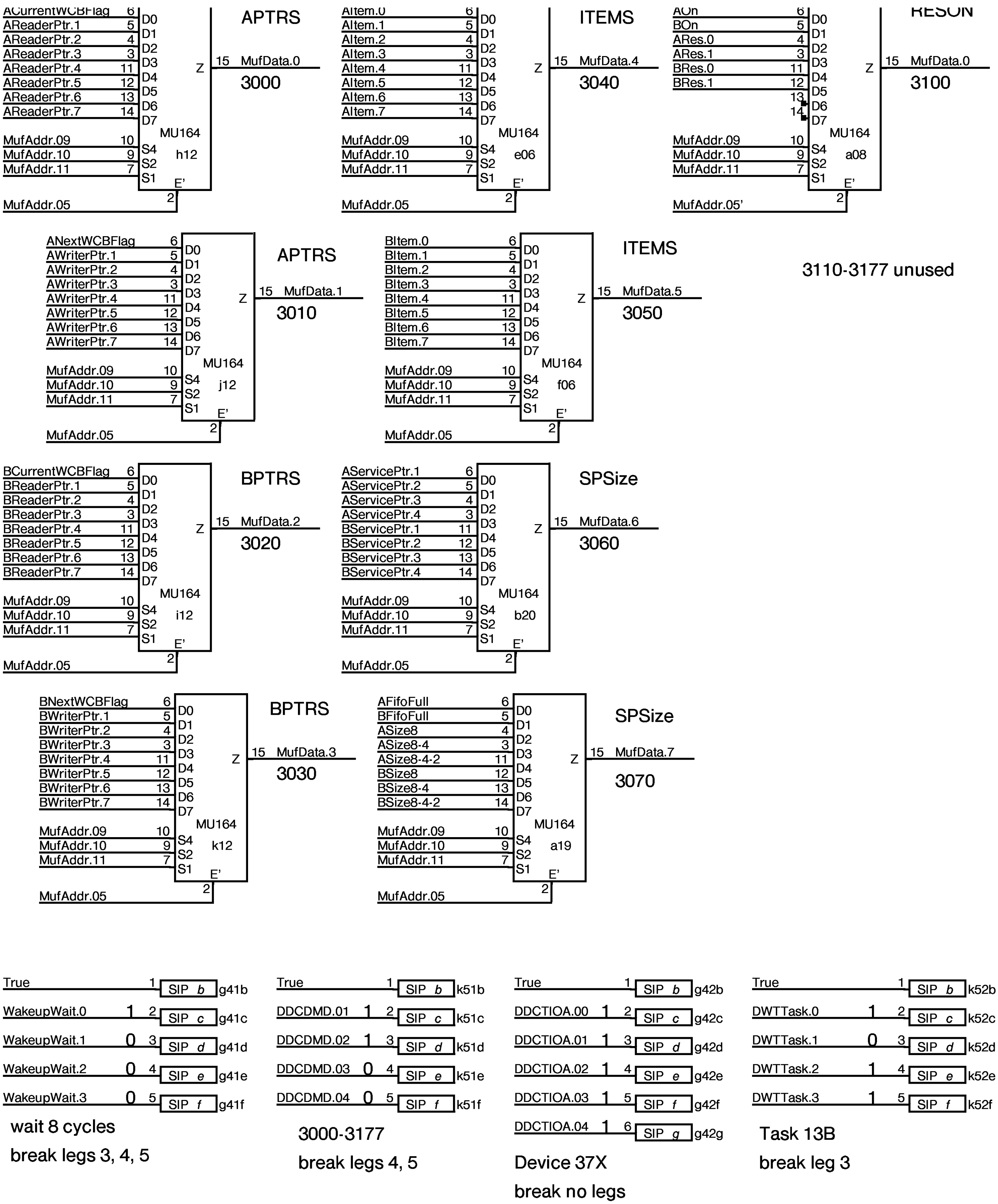
CBlank must be phased so that it changes when HRamAddr.10 goes from 1 to 0. However, CBlank is delayed by two pixel clocks from RamdHBlank, so the HRam must be programmed to make RamdHBlank change on the 0 to 1 transition of HRamAddr.10. This is slightly confusing, so be careful.



XEROX PARC	Project		File	Designer	Rev	Date	Page
	Dorado	DDC OIS Terminal Interface	DispY21.sil	K. Pier	Ci	10/29/79	21

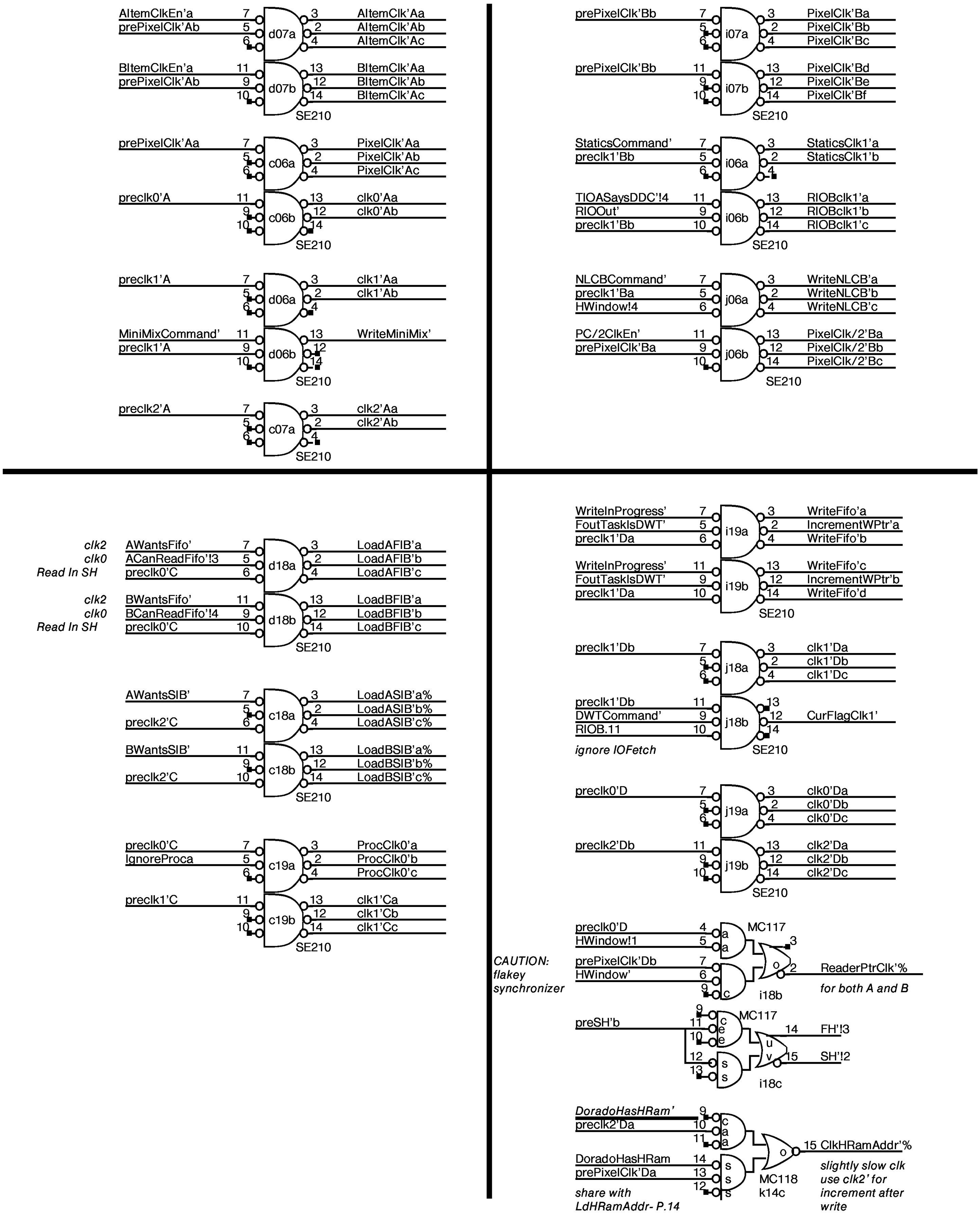


XEROX PARC	Project		File	Designer	Rev	Date	Page
	Dorado	DDC Statics Register and Muffler Control	DispY22.sil	K. Pier	Ci	7/19/79	22

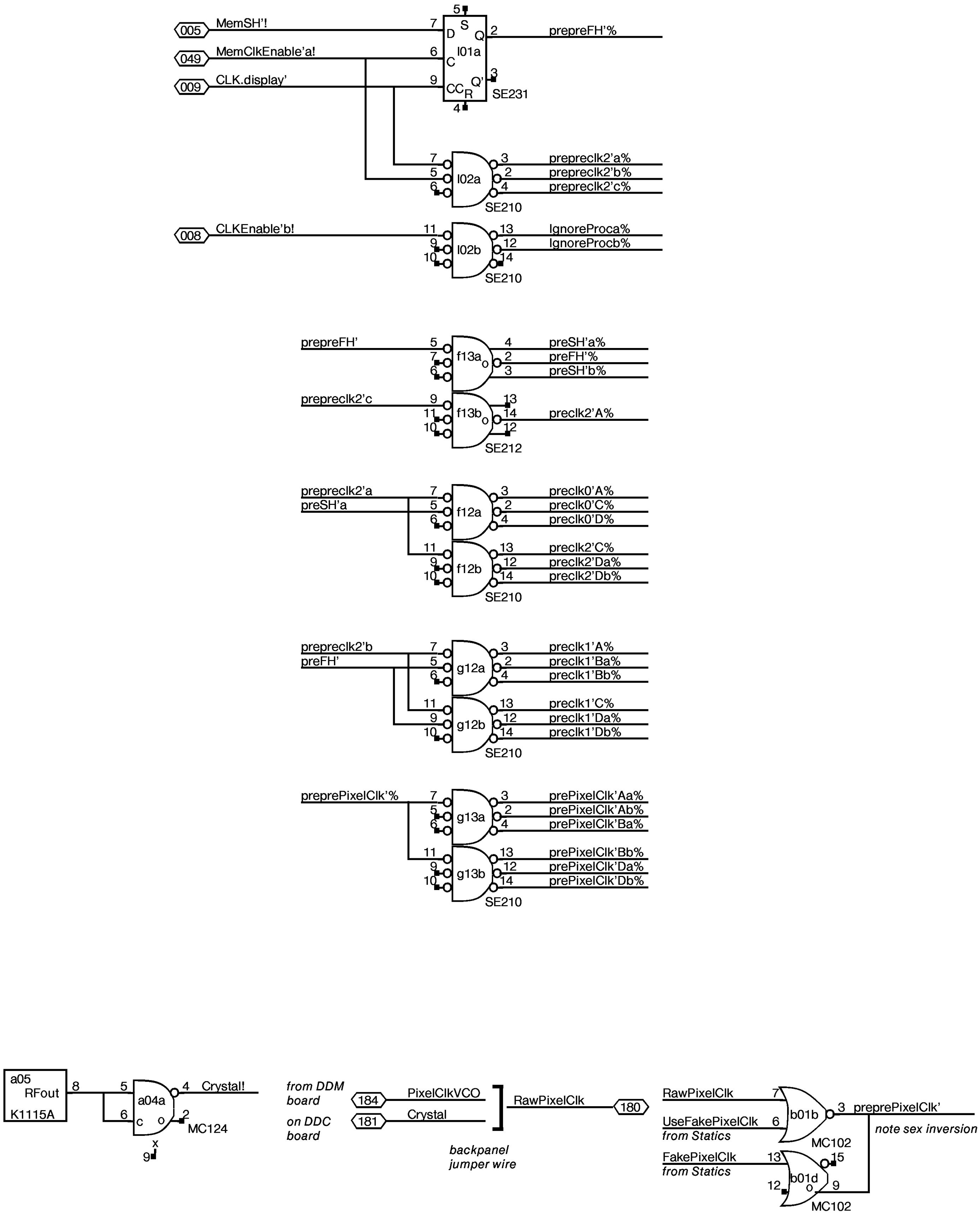


All SIPs are standard 100 ohm terminators with legs broken as shown

XEROX PARC	Project		File	Designer	Rev	Date	Page
	Dorado	DDC Mufflers and SIPs	DispY23.sil	K. Pier	Ci	7/19/79	23



XEROX PARC	Project Dorado	DDC Clock Drivers	File DispY24.sil	Designer K. Pier	Rev Ci	Date 11/29/79	Page 24


MemSH'!
MemClkEnable'a!
CLK.display'
l01a
SE231
prepreFH'%
l02a
SE210
prepreclk2'a%
prepreclk2'b%
prepreclk2'c%
CLKEnable'b!
l02b
IgnoreProca%
IgnoreProcb%
prepreFH'
f13a
preSH'a%
preFH'%
preSH'b%
prepreclk2'c
f13b
SE212
preclk2'A%
prepreclk2'a
preSH'a
f12a
f12b
preclk0'A%
preclk0'C%
preclk0'D%
preclk2'C%
preclk2'Da%
preclk2'Db%
prepreclk2'b
preFH'
g12a
g12b
preclk1'A%
preclk1'Ba%
preclk1'Bb%
preclk1'C%
preclk1'Da%
preclk1'Db%
preprePixelClk'%
g13a
g13b
prePixelClk'Aa%
prePixelClk'Ab%
prePixelClk'Ba%
prePixelClk'Bb%
prePixelClk'Da%
prePixelClk'Db%
a05
RFout
K1115A
a04a
Crystal!
MC124
from DDM board
on DDC board
PixelClkVCO
Crystal
RawPixelClk
backpanel jumper wire
RawPixelClk
UseFakePixelClk
from Statics
FakePixelClk
from Statics
b01b
b01d
MC102
preprePixelClk'
note sex inversion

CONNECTOR CONNECTOR TIOA TW CONNECTOR IOBLo CONNECTOR IOBHi CLK

A	B>Ct a 181	b 168	c 153	d 137	e 124	f 109	93 g	80 h	64 i	48 j	33 k	20 l	B
1 41	DAC Plat	PCLK B>Ct 102		TIOA DEF 176 1	TIOA 176 1	DHTTw RdFifo 135	ALTO 74128	ALTO 125	RIOB 176 rc1	RIOB 176 rc1	RIOB 176 rc1	SE231	1
2	DAC 10318			OIS 176	TIOA = D 113	MMMux 158	MMMux 158	NLCB 145 W	DIOB 100	NLCB 145 W	NlcbAd F16 P2	SE210	2
3 42	DAC Plat	MMix F16 PC	OIS ICEn 176 PC	DWTTw 231 0	TIOA = D 113	ALTO 105	NLCB 145 W	CURSR 141 P	CURSR 141 P	CURSR 141 P	NlcbAd 197	CURSR 141 P	3
4	PCLK 124 BCD	MMix F16 PC	OIS 105	OIS 101	OIS 176	MMix	ClcbEn 161	VSyC 104	CURX F16 P	CURX F16 P	CURX F16 P	CURX 103	4
5 43	PCLK Xtal	LSR 104	OIS F00	OIS F16	OIS 174	MMix	ClcbEn 161	STAT 231 sc1	STAT 176 sc1		195 RIOB'	CURSR 104	5
6	AdICEN 121	BdICEn 121	PC MMix	0,1	MU AITEM	MU BITEM	ASIZE F16 P2	ARES F16 P2	sc1 rc1	NLCB 1 PC/2		MIDAS 158	6
7 44	ARESC F16 P	BRESC F16 P	2 210 B	ABITEMCLK	AITEM 104	BITEM 104	BSIZE F16 P2	BRES F16 P2	PC PC		PULSE 195	MIDAS 158	7
8	MU ABOFF OIS RES	LSR 118	ASRC F16 I	BSRC F16 I	AITEM 104	BITEM 104	HWNDO 135 P2	BWDTH F16 P	BWDTH F16 P	BWDTH F16 P	BWDTH 103	MIDAS 164	8
9 45	ASYNC 231 2	ASYNC 231 2	ASYNC 105	BSYNC 231 2	ASYNC 135 I	BSYNC 135 I	MODE F16 P2	ABOFF 135 P	AWDTH F16 P	AWDTH F16 P	AWDTH F16 P	176 PC P	9
10	ASR I F00	BSR I F00	ASR F00	BSR I F00	ASR I F00	BSYNC 105	VCW F16 P2	BLMarg F16 P	BLMarg F16 P	BLMarg F16 P	BLMarg 105	HRmOut F16 P	10
11 46	ASR I F00	BSR I F00	ASR I F00	BSR I F00	BSR I F00	BSR I F00	WANT 117	ALMarg F16 P	ALMarg F16 P	ALMarg F16 P	ALMarg 105	AWDTH 103	11
12	ASR I F00	BSR I F00	ASR I F00	BSR I F00	ASR I F00	pc0' pc2' SE210	pc1' pc1' SE210	MU ARP AF	MU BRP BF	MU AWP AF	MU BWP BF		12
13 47	ASIB 176	BSIB 176	AFIB F00	BFIB F00	BFIB F00	pc2 FHSH SE212	ppc' ppc' SE210	HRAddr F16 H	HRAddr F16 H	HRAddr F16 H	HRAddr 103	MIDAS 166	13
14	ASIB 176	BSIB 176	AFIB F00	AFIB F00	BFIB F00	COMM' 176 1	flags 117	HRam 415 HW	HRam 415 HW	HRam 415 HW	HRamC 118	HRamC 118	14
15 48	ASIB 176	BSIB 176	AFIB F00	BFIB F00	IP	NFlags 231 1	CFlags 231 cf	DHasHR 231 1,p2	AFULL 149	BFULL 176 1	AFULL 176 1	BFULL 149	15
16	ASIB 176	BSIB 176	AFIB F00	BFIB F00	IP	IP			ARP rpc F16	BRP rpc F16	AWP wpc F16	BWP wpc F16	16
17 49	ASIB 176	BSIB 176	AFIB F00	BFIB F00	IP	IP	FIFO	FIFO	ARP rpc F16	BRP rpc F16	AWP wpc F16	BWP wpc F16	17
18	ASIB 176	BSIB 176	ABSIB 2 ABSIB 2	ABFIB 0	IP	IP			FHSH rpc 0 117	1, cflg 1	FAM 174	PTRS 103	18
19 50	MU BA-SZE	ABSize 103	PROC0' 1'		IP	IP	FIFO	FIFO	fifo 1 wpc 1	2 0	FAM 174	Sync 176	19
20		MU SP	AFIB F00	BFIB F00	IP	IP			FA 2 231	FAM FH 174 SH	FAM 174	IncSP 103 C	20
21 51			AFIB F00	BFIB F00	IP	IP	FIFO	FIFO	FA 2 231	FA 2 231			21
22	CTDWT 117 C		OIS 231	IP	IP	IP			FA 2 231	d 195	MIDAS 176	MIDAS 176	22
23 52	102 C	clk0 0 176 FG	GETS 1 231	BLKD B 1 135	ASP 1 F16	BSP 1 F16	FIFO	FIFO	FOUT 1 176	FOUT 1 176	FOUT 1 176	MIDAS 102 BC	23
24	NEXT 113	SUBT 106 A	105	195	WSpace F16 0	HELD 231 0	IOInOut 1660	TIOA = D 161	FOUT 0 176	FOUT 0 176	FOUT 0 176	FTsk 113	24
C	a 11	b 26	c 39	d 55	e 70	Hold f 86	99 g	114 h	129 i	143 j	159 k	174 l	D

E NEXT SubT BLK FIN IOin/out CONNECTOR IOR FNXT FOUT FTsk DMux E

XEROX PARC	Project	Reference	File	Designer	Rev	Date	Page
	Dorado	DDC Board Layout	DispY26.sil	K. Pier	Ci	3/26/81	26

DEVICE	TIOA	I/0	TASK	FORMAT and COMMENTS
STATICS	377	O	DHT, EMU	See Below
NLCB	376	O	DHT, EMU	NLCBAddr.0-3,,Data.4-15
HRAM	375	O	DHT, EMU	Keep,Write,LoadAddr,0,Data.4-15
DHTFLAG	374	O	DHT	....,ANextWCBFlagSet,BNextWCBFlagSet,0 bits 13,14,15
DWT	373	O	DWT	IOFetch signal,....,Set/Clr CWCBFlagANDClrNWCBFlag bit 11, ... , bit 15
MiniMixer	372	O	DHT, EMU	address,,data always writes MiniMixer RAM
STATUS	370	I	DHT, EMU	Selected muffler input returned in bit 15
PIXELCLK	367	O	DHT, EMU	Pixel clock rate located on mixer board
MIXER	366	O	DHT, EMU	Keep,Write,LoadAddr,0,Data.4-15 located on mixer board

00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	
X	Muff Addr .05	Muff Addr .06	Muff Addr .07	Muff Addr .08	Muff Addr .09	Muff Addr .10	Muff Addr .11	XSync Enable	Fake Pixel Clock	Use Fake Pixel Clock	Unused	Unused	Unused	DWT Shut Up	DHT Shut Up-Reset DDC	Statics
NLCB Addr .00	NLCB Addr .01	NLCB Addr .02	NLCB Addr .03	NLCB Data 12 bits												NLCB
Keep HRam'	Write HRam'	Load Hram Addr				Addr.0	Addr.1	Addr.2	Addr.3	Addr.4	Addr.5	Addr.6	Addr.7	Addr.8	Addr.9	HRAM
													HSync	HBlank	Half Line	
													Set BNext WCB Flag	Set ANext WCB Flag	Must Be 0	DHTFLAG
											IOFetch signal	Must Be 0	Must Be 0	Must Be 0	Set/ Clr Cur WCB Flag	DWT
Address.0-7								Data.0-7								MiniMixer
								ClkRate.0-7								PIXELCLK
Keep Mixer'	Write Mixer'	Load Mixer Addr	X		Addr.0	Addr.1	Addr.2	Addr.3	Addr.4	Addr.5	Addr.6	Addr.7	Addr.8	Addr.9	Hi/Lo select	MIXER
				Mixer Data 12 bits												

DDC Slow IO System

Next Line Control Block Format

Address	Name	Format
0	VCW vertical control word	0...0, VBlank, VSync, EvenField
1	AMargin	LMarg[00..11]
2	AWidth	Width[00..11]
3	AFifoAddr	FifoAddr[0..7] *must be even
4	AScan	0...0,Polarity,Resolution[0..1],Size8,Size4,Size2,Size1
5	ModeControl	0..0,24Bit,AByPass,BByPass,A8B2
6	reserved	
7	reserved	
10	unused	
11	BMargin	LMarg[00..11]
12	BWidth	Width[00..11]
13	BFifoAddr	FifoAddr[0..7] *must be even
14	BScan	0...0,Polarity,Resolution[0..1],Size8,Size4,Size2,Size1
15	CursorX	CursorXCount[0..11]
16	CursorLo	CursorLoByte[4..11]
17	CursorHi	CursorHiByte[4..11]

DDC	SIZE	Pin	DDM	
AItem	8	xxx	AItem	85-88-89-92-93-96-97-100
BItem	8	xxx	BItem	101-104-105-108-109-112-113-116
CursorData	1	117	CursorData	
AItemClkEn	1	140	AItemClkEn	
BItemClkEn	1	141	BItemClkEn	
AOff	1	048	AOff	
BOff	1	045	BOff	
HSync	1	041	HSync	
HBlank	1	080	HBlank	
HalfLine	1	081	HalfLine	
VSync	1	044	VSync	
VBlank	1	084	VBlank	
PixelClk	1	184	PixelClk	
Crystal	1	181	Crystal	
Oscillator	1	180	Oscillator	
Modes	4	xxx	Modes	28-29-32-33
XHSync	1	036	XHSync	
XVSync	1	037	XVSync	
XSyncEn	1	040	XSyncEn	
OISData.0	1	144		7 wire interface cable
OISData.0’	1	145		
OISData.1	1	148		
OISData.1’	1	149		
OISData.2	1	152		
OISData.2’	1	153		
OISData.3	1	156		
OISData.3’	1	157		
OISClkA	1	161		
OISClkA’	1	160		
OISClkB	1	164		
OISClkB’	1	165		
OISDataOut	1	169		
OISDataOut’	1	168		
AltoVideo	1	072		Alto monitor cable
AltoHSync	1	073		
AltoVSync	1	076		
AltoCSync	1	077		Grey level driver cables 2 coax
DAC	1	172		
DAC GND	1	173		
spare	1	176		
KeyboardData	1	177		from Baseboard

Seven Wire Interface Cable

Signal	DA15P connector	Wire	Dorado backpanel	Signal
OISData.0	7	1	144	OISData.0
OISData.0’	8	1	145	OISData.0’
OISData.1	5	1	148	OISData.1
OISData.1’	6	1	149	OISData.1’
OISData.2	3	1	152	OISData.2
OISData.2’	4	1	153	OISData.2’
OISData.3	1	1	156	OISData.3
OISData.3’	2	1	157	OISData.3’
OISClkA	9	1	161	OISClkA
OISClkA’	10	1	160	OISClkA’
OISClkB	11	1	164	OISClkB
OISClkB’	12	1	165	OISClkB’
OISDataOut	13	1	169	OISData
OISDataOut’	14	1	168	OISData’

DispY

BASEBOARD

From	Wire	To (BASEBOARD)
169	*	169
168	*	168
DispY 177	KeyboardData *	177

* indicates backpanel wiring

Grey level cabling

BNC connector		Backpanel	Signal	
BNC connector	coax center	172	DAC output	DispY
	coax shield	173	DAC ground Insulated from chassis ground	
BNC connector	coax center	077	CompSync	DispY
	coax shield	078	ground	

Note: CompSync currently appears in the Alto cable. The Alto display doesn’t make use of CompSync. We could simply replace the CompSync pair in the Alto cable with the coax. We then have the capability to attach a local Alto type display and a remote display.

Six BNC connectors mounted on the "square" side. Cabling not yet defined.

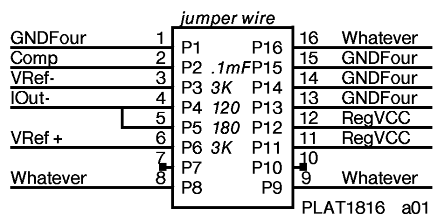
1. Plug an MC10318 D/A converter into location a02 .
2. Plat1816 in locations a01 and b02 are discrete components, shown below.
3. SIPs are 100 ohm terminator package with legs broken:

location	break legs
g41	3,4,5
g42	none
k51	4,5
k52	3

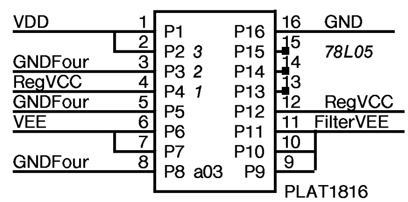
See page 23.

4. SIPs at locations d42 and e42 are 220 ohm value instead of 100 ohm.

5. Crystal oscillator K1115A, location a05, value 20 MHz.



Platform is DAC components



Voltage filters

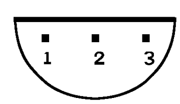
1	jumper wire	16
2	0.1mFarad	15
3	3 KOhm	14
4	120 Ohm	13
5	180 Ohm	12
6	3 KOhm	11
7		10
8		09

location a01

1	0.1mFarad	16
2 in 3	78L05	15
3 gnd 2		14
4 out 1		13
5	0.1mFarad	12
6	12 mHnry	11
7	2 ohm	10
8	+ 22 mF	09

location a03

See Page 20.



1. Output
2. Ground
3. Input

Bottom view of pinout for 78L05